

Teaching Portfolio

of

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Supplementary materials can be found at: <https://dearmahmud.github.io>

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Preface

This teaching portfolio shows how I support student learning, test new teaching approaches, develop as an educator, and contribute to teaching quality within my department and beyond. I have organized it to show a clear link between what I believe about teaching and what I actually do in practice. It begins with a short biography, then presents my teaching philosophy, followed by the teaching and assessment methods that shape my work. It also explains how these ideas inform my mentoring, curriculum design, and academic leadership.

This portfolio reflects a seventeen-year journey guided by one lasting goal: helping students grow into professionals and engineers who can contribute to a better world. One guiding principle has defined how I teach and why I teach: *effective learning begins when we can enter the minds of students and ignite the spark that drives their curiosity*. My role, as I see it, is to create the conditions in which that curiosity can grow. For that reason, I use teaching methods that are practical, flexible, and responsive to the students I teach.

The sections that follow show how this philosophy appears in my course design, teaching practice, assessment, learning outcomes, and mentorship. Together, they reflect my effort to help students move from understanding ideas in theory to applying them with confidence, independence, and professional judgment.

Biography

I am Mohammad Mahmudul Hasan, an Associate Professor in the Department of Electrical and Electronic Engineering (EEE) at the University of Information Technology & Sciences (UITS), Bangladesh.¹ I have more than 17 years of experience in teaching, research, and student supervision at undergraduate and postgraduate levels across Bangladesh, India, Japan, the Czech Republic, and Norway.

I completed my bachelor's degree (2004–2008) and master's degree (2008–2010) at KIIT University, India, and earned my PhD (2022–2025) from the Norwegian University of Science and Technology (NTNU), Norway. My PhD thesis was nominated for the *Best Doctoral Thesis Award* at NTNU in 2025. Earlier, I received double *Gold Medals* for academic performance during my master's studies at KIIT University in 2010, along with the *Chancellor's Merit Scholarship* for academic excellence throughout my bachelor's and master's studies (2004–2010). In 2024, I was a Visiting Researcher at Brno University of Technology (VUT), Czech Republic.

I have been affiliated with UITS since 2011. I served as Assistant Professor from 2011 to 2025 and have been Associate Professor since 2025. During this time, I also held several academic leadership roles, including Head of the Department (2019–2022), Director of the Information and Communication Technology (ICT) Cell (2020–2022), and Advisor to the ICT Cell (2018–2022). I also served in institutional roles including Member of the Finance Committee (2020–2022), Editorial Board Member (2020–2022), Member of the Academic Council (2019–2022), and Member Secretary of the Faculty (2018–2022). Alongside my academic work, I worked with industry as a consultant and technical trainer for Sanyo Engineering & Construction Inc. (Suntec Group, Japan) during 2018–2019.

I have also contributed to several funded research and institutional projects. These include *Machine Learning Assisted Signal Processing for Antenna-Sensor Arrays* as a Work Package Member (2024–2025), *Air Quality Monitoring Exploiting Massive Multiple-Input Multiple-Output Antennas* as a Work Package Member (2022–2025), the *University Comprehensive Academic Management, Software Implementation Project* as Project Manager (2020–2022), and the *Higher Education Quality Enhancement Project* as a Member (2017–2018).

Teaching Philosophy

I have been teaching at the university level since 2009. Over time, I have often reflected on a difficult question: *why are so many of the world's most successful technology entrepreneurs' college dropouts?* They entered university to learn, yet many found little space there for the kind of learning that worked for them. Instead, they turned to experimentation and learned by *doing*. That contrast stayed with me. It made clear that students do not learn in the same way. Some do well in structured, lecture-based settings. Others need a more active and flexible approach. The issue is often not their ability, but whether our teaching methods are broad enough to reach them. This has shaped my commitment to building classrooms where more students can participate, engage, and succeed.

In my early years, my own teaching was mostly traditional and lecture-based. I focused on covering the curriculum and making sure students understood the core concepts of the discipline. That changed with experience. Classroom realities, student feedback, and pedagogical training pushed me to rethink what good teaching requires. My philosophy gradually shifted from delivering information to helping students develop critical thinking, adaptability, and a lasting interest in learning. My mentors influenced that shift in important ways. Some taught me the value of precision and intellectual discipline. Others showed me that students learn not only from what a teacher says, but also from how a teacher thinks, questions, and responds. I also learned from senior collaborators that difficult ideas become more accessible when they are presented clearly and connected to a meaningful context. One lesson has stayed with me throughout: there is no single best way to teach. The right method depends on the subject, the context, and the students in the room.

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I do not believe one fixed teaching method can serve every class well. Students learn differently, and different topics demand different forms of engagement. My experience has shown that relying on one strategy usually leaves part of the class behind. Good teaching requires adjustment, to the subject, the task, and the group itself. Sometimes that adjustment has to happen during the class, in response to where students hesitate, how they reason, and what they already understand. The goal, however, remains the same: to create conditions in which students think critically, solve problems, and justify their decisions. Real learning is active. It develops when students work through difficulty, test ideas, defend choices, and refine their understanding through evidence and reflection.

Although my teaching adapts to each group of learners, it is grounded in three core principles: (a) fostering a safe and positive learning environment, (b) committing to a student-centered and active learning model, and (c) using collaborative learning as a meaningful part of the educational process.

A Safe and Positive Learning Environment

Effective learning depends on a classroom environment in which students feel safe enough to participate, question, and take intellectual risks. In my experience, students engage more deeply when they know that uncertainty, hesitation, and early mistakes are part of learning rather than signs of failure. A safe learning environment does not remove challenge. It makes challenge possible. When students are not afraid of embarrassment or dismissal, they are more willing to test ideas, ask for clarification, and stay engaged with difficult material. That sense of security supports every collaborative activity I design.

Research by Thompson and Wheeler 2008 suggests that intellectual risk-taking depends on psychological safety. Young, Williamson, and Egan 2015 also shows that students are more willing to ask questions, engage in discussion, and test ideas when the classroom feels safe and supportive. This aligns closely with my own practice. I try to treat the classroom as a space for dialogue rather than one-way delivery. I want students to see that hesitation is acceptable, disagreement can be productive, and mistakes can lead to deeper understanding. That requires trust, and trust has to be built deliberately.

Research also shows that the learning climate influences student behavior, achievement, and satisfaction (Genn 2001a and Genn 2001b). Hutchinson 2003 describes a safe environment as one in which students can identify gaps in their knowledge, test their limits, and experiment with ideas. I have found the same. Students learn more effectively when new ideas are connected to what they already know and when the classroom gives them room to build understanding step by step. Confidence grows gradually. So does learning.

I first understood this clearly when I began teaching at KIIT University in 2009. I had just completed my master's degree and was close in age to many of my students. That created an unusual dynamic. I was expected to guide them academically while still adjusting to my own new role as a teacher. My first priority was to build mutual respect so that students felt comfortable approaching me, speaking openly, and engaging seriously in class. That experience showed me that psychological safety begins with trust and that trust often begins with how a teacher relates to students.

Later, at UITS, I faced a more complex classroom. I often taught two distinct groups together: recent secondary-level graduates from varied educational backgrounds, and diploma engineers, often aged 40 to 60, returning to academic study after many years in professional life. Many of the diploma engineers brought strong practical experience but felt uncertain about forgotten foundational material and hesitant to speak in front of younger students. Creating a safe environment for both groups required careful effort. I encouraged the senior students to treat their field experience as a valuable academic resource, and I encouraged the younger students to support them with coursework and study routines. Over time, the gap between the groups narrowed. The class became more cooperative, and students began learning from one another in meaningful ways. In several cases, younger students later secured industrial internships through workplace connections shared by the diploma engineers.

That outcome showed me that a supportive classroom can strengthen both learning and professional development.

A safe learning environment also depends on everyday practices. I try to learn students' names quickly because recognition matters. It helps students feel seen and respected. I set clear expectations for openness, civility, and mutual respect from the beginning of the course, and I reinforce those expectations throughout the semester. I also design activities such as small-group discussions, case-based tasks, and reflective writing to lower the barrier to participation. These activities help students do more than practice course content. They help students develop confidence in expressing their own thinking.

Studies (Clapper 2010; Thompson and Wheeler 2008; Young, Williamson, and Egan 2015) emphasize the importance of trust and mutual respect in supporting active participation. My own experience at KIIT and UITS confirms this. Regardless of age, background, or prior preparation, students carry insecurities into the classroom. If those insecurities are ignored, learning suffers. If they are addressed with care, students participate more fully and learn more confidently.

Student-Centered Active Learning

I believe students are responsible for their own learning, and teachers are responsible for designing the conditions that make that learning possible. Over seventeen years of teaching, this belief has grown from experience rather than theory. Early in my career, I thought clear and thorough lectures were enough. But while teaching embedded circuits in 2009, I saw a repeated problem. Students could follow a circuit diagram, repeat definitions, and solve standard textbook questions, yet still struggle to build a working prototype on a breadboard or deal with a sensor that behaved unpredictably. The gap was obvious. They understood the theory at one level, but they could not always use it. That experience convinced me that engineering competence develops through active work, not through listening alone. Students need to design, build, test, fail, and troubleshoot.

That realization changed how I approached teaching. I began experimenting with different instructional models, including traditional lectures, guided problem-solving, scaffolded demonstrations, and slide-based content delivery (Maor et al. 2025; J. Eison 1991; Capone 2022). Each method helped some students, but none worked equally well for the whole class. I gradually moved toward approaches that placed students at the center of the learning process, especially *problem-based learning*, where students learn by engaging with complex and realistic problems. That fit naturally with engineering education. I also drew on active learning principles, which argue that students must do something with knowledge, discuss, solve, test, compare, or create, if they are to understand it fully (Biggs and Tang 2011; Armbruster et al. 2009; Al Abri and Al-Mekhlafi 2024).

While teaching at KIIT University, India (2009–2010), I began applying these ideas in more deliberate ways. My earliest efforts focused on final-year capstone teams. I reorganized the opening weeks of the semester into an intensive design sprint where teams defined problems, assessed risks, held daily stand-up meetings, and produced early prototypes. I reduced formal lecturing to short micro-lectures and stepped back so that students had to take ownership of the process. The result was immediate. Teams delivered workable prototypes in 12 weeks instead of 24, and several students moved directly into industry based on the quality of those projects. That experience showed me how quickly learning accelerates when students must make decisions, justify them, and defend them in front of others.

I extended the same approach into laboratory teaching. Circuits and instrumentation labs were redesigned as problem-based sessions rather than exercises built around fixed instructions. Instead of asking students to reproduce known outcomes, I gave them open engineering tasks such as designing low-cost sensing boards, troubleshooting faulty filters, or reducing system noise. They had to identify constraints, test alternatives, compare results, and explain their choices. Their reports improved. They became more analytical, more evidence-based, and less dependent on copied formulas. At that

point, I stopped seeing active learning as a useful addition to a course. It had to become part of the course structure itself.

One example that shaped my thinking came from a unit on analog-to-digital conversion. Rather than starting with explanation, I gave student teams a potentiometer, a microcontroller, and an LED, then asked them to make the LED brightness change smoothly with the position of the knob. To do this, they had to read datasheets, configure registers, interpret unstable voltage readings, and solve practical problems together. Only after that struggle did I lecture on the underlying principles. The difference was clear. Students listened with attention because they were now looking for answers to problems they had already encountered. Their curiosity had a real source. In that sense, the lecture became more effective because active work came first. This closely matched the argument made by Bonwell and J. A. Eison 1991, that knowledge is built through engagement rather than passive reception.

Inspired by educators such as Mazur 2009; Prince 2004; Olewnik, Chang, and Su 2023, I also began integrating smaller active-learning elements into otherwise traditional course structures. In embedded systems courses, for example, I would stop during a lecture and ask students to design a short microcontroller routine in pairs. At first, many resisted because listening felt safer than attempting an answer in public. But once they began discussing register settings, timing constraints, and implementation errors with one another, they started to develop engineering intuition in a much more practical way. Later, I introduced structured exercises such as debugging races, where teams competed to identify faults in deliberately flawed code. These activities changed the atmosphere in the room. Students who had previously remained silent started contributing, and complex concepts became easier to teach because they were already thinking through the material while learning it.

In communication engineering courses, I use practical tasks to connect theoretical concepts with real systems. Students calculate the link budget for a mobile communication network serving their campus, compare signal strength measurements across different locations using smartphone applications, and analyze coverage and handover behavior while moving between base stations. They also design wireless communication links, evaluate Wi-Fi performance under different environmental conditions, and examine how antenna placement affects coverage and reception.

To strengthen foundational concepts, students build and test AM and FM modulation and demodulation circuits, observe signal waveforms on oscilloscopes, and explain the operating principles to their peers. In project-based work, they develop simple radio communication systems and investigate how noise, interference, and propagation conditions affect system performance. These activities help students connect abstract theory to engineering practice while strengthening technical understanding, teamwork, and problem-solving.

I also introduced industry-based case assignments to bring professional judgment into the classroom. Diploma engineers in my classes sometimes brought real technical failures from their workplaces, and student teams worked together to diagnose the issues, choose models, and justify possible solutions. Since these cases did not have one correct answer, I assessed the quality of the reasoning rather than whether students reproduced a predetermined result. I looked at how clearly they defined the problem, how carefully they chose assumptions, and how well they balanced trade-offs. This shifted their attention away from memorization and toward professional thinking. In several cases, internships followed directly from these collaborations.

When I moved to UITS, I continued refining this approach. I added short reflection notes, quick in-class design challenges, and peer-teaching activities. In courses such as Digital Signal Processing, I used a flipped classroom model in which students reviewed short content blocks before class, allowing class time to be used for challenge problems and conceptual diagnostics in MATLAB worked on in pairs (Al Abri and Al-Mekhlafi 2024). This changed the function of class time. Instead of passively receiving information, students used it. I also introduced case-based learning around practical problems such as noise reduction in communication systems. This encouraged critical thinking and reduced students' fear of making mistakes because the emphasis was on reasoning

through the problem rather than producing an immediate perfect answer (Das et al. 2021; Kaur et al. 2020).

One of the clearest examples of the value of active learning came from a professional training program for newly graduated engineers recruited by a Japanese company to work on overhead transmission line construction. They had strong theoretical knowledge, but they were not yet ready for the realities of high-voltage work. That kind of work is technically demanding and potentially dangerous, so classroom explanation alone was not enough. I took them to an operating substation and guided them through the equipment, safety procedures, and field practices. The change in understanding was immediate. At the end of the visit, one participant told me, “We learned more about this in a single day than we did in four years of engineering.” I have never forgotten that remark. It captured something essential. When students engage directly with authentic tasks in real settings, theory becomes clearer, more useful, and much harder to forget.

Collaborative Learning

Collaborative learning is a central part of my teaching because it reflects how engineering work is actually done. Engineers rarely solve complex problems alone. They work in teams, combine different kinds of expertise, and explain technical ideas clearly to others. I treat collaborative learning not only as a teaching strategy but also as preparation for professional practice. As noted by Gokhale 1995a, collaborative learning develops through interaction, dialogue, and feedback between students and teachers. Research also shows that students working in cooperative teams often reach deeper levels of understanding, retain knowledge longer, and strengthen critical thinking through shared problem-solving (Yang 2023; Laal and Ghodsi 2012; Ferreira, Zabolotna, and Lee 2025).

In my classes, I use small-group work regularly because it helps students learn how to reason with others, not just complete a task. Research supports this approach. D. W. Johnson and R. T. Johnson 1999 found that cooperative learning can improve cognitive engagement and knowledge retention compared with isolated work. I have seen the same in practice. Students who are quiet in a full class often contribute meaningfully in small groups. Once they begin explaining their thinking, questioning assumptions, and responding to peers, their understanding becomes clearer and more confident.

I use collaborative tasks in both theoretical and applied courses. In a computer networks course, for example, student teams design a simplified local area network for a specific region. They analyze traffic requirements, propose a network structure, and justify their choices in terms of cost, scalability, and reliability. In a control systems course, students work together to model and simulate a feedback control loop, dividing responsibility across parameter tuning, stability analysis, and documentation. These tasks require students to combine theory with practical judgment while learning how to negotiate different technical viewpoints (Gokhale 1995b; Laal and Ghodsi 2012; Li 2025).

Laboratory teaching gives especially strong opportunities for collaborative learning. I usually organize students into small teams with shared but distinct roles such as circuit assembly, measurement, observation, and data interpretation. In a high-voltage laboratory, for instance, students test insulation breakdown under different conditions and must make decisions together about procedure, safety, and interpretation of results. This improves technical accuracy, but just as important, it develops communication and decision-making under real constraints. In project-based courses, I also ask students to use digital tools such as Microsoft Teams or Google Docs so that collaboration continues beyond the classroom.

Projects are often where collaborative learning becomes most effective. In class projects and final-year design work, students engage in a sustained process of joint problem definition, experimentation, discussion, and revision toward a shared goal (Gokhale 1995b). This process builds more than technical skill. It also helps form a learning community, reduces anxiety, and encourages critical thinking (Laal and Ghodsi 2012). I have repeatedly seen that when students work well together, the progress of one

student often supports the progress of others.

For that reason, I often assign teams randomly, and I do this deliberately (Bansal et al. 2019). I do not want group formation to depend only on friendship or familiarity. In engineering practice, people rarely choose their collaborators. They are expected to contribute effectively in diverse teams, including teams that are not ideal. By mixing students intentionally, I prepare them for the technical and interpersonal realities of industry and research environments (Bansal et al., 2019). The point is not to create comfortable teams. It is to help students learn how to work productively in real ones.

This approach became especially valuable in my diverse classrooms at UITS, where younger graduates and senior diploma engineers often studied together. I used collaborative learning to turn that difference into a strength. Senior learners brought field experience and practical judgment, while younger students often contributed stronger digital and software skills. In one project on IoT-based energy monitoring, mixed teams designed prototypes using Arduino boards. The work improved technical understanding, but it also built confidence and mutual respect across age and background (Abramczyk and Jurkowski 2020). In several cases, younger students later secured industrial internships through contacts provided by the senior learners.

I also use structured peer review to strengthen collaboration. In project-based coursework, student teams design, simulate, and validate large circuits, then present their architectural plans and testing procedures to other groups for critique. They receive feedback on technical choices, methodology, compliance with specifications, and implementation quality. This improves the final design, but it also teaches students how to give and receive technical criticism in a professional way.

In analytically demanding subjects, I use a structured problem-solving format similar to a modified Think-Pair-Share model (Al Abri and Al-Mekhlafi 2024; Guenther and Abbott 2024). I introduce a difficult problem, such as stability analysis or filter design, and ask students to begin individually. They then compare approaches in pairs, resolve differences, and finally work in larger groups to develop and present a stronger solution to the class. This structure keeps each student accountable while using the group to deepen understanding of difficult concepts.

Most assignments in my courses are group-based because collaborative problem-solving mirrors engineering practice. When students debate the assumptions behind a propagation model, compare vector network analyzer readings, or design a compact microwave resonator together, they sharpen their thinking more effectively than they do alone. They must justify choices, test weak arguments, and integrate useful ideas from others. Those are core habits of technical work.

The results are clear in both student performance and feedback. Students retain theoretical concepts better, their project work becomes more original, and the classroom becomes more of a learning community than a teacher-centered hierarchy. The transition is not always easy. Some students initially resist the added responsibility and cognitive demand. But over time, most begin to see the value. Their final work shows stronger reasoning, fewer technical errors, and more independence. They move beyond repeating examples and start designing with confidence. That is the outcome I want from collaborative learning.

Teaching and Assessment Repertoire

Assessment is a core part of how I teach because it shapes what students attend to, how they practice, and how they improve. I design teaching and assessment together so that each method supports the intended learning outcomes of the course. My approach is guided by the Washington Accord², which emphasizes internationally recognized graduate competencies and outcome-based curriculum design (Aldridge 2012; Rathy, Sivasankar, and Gnanasambandhan 2020; Rao and Rao n.d.). That framework helps me plan courses in a way that connects teaching, assessment, and graduate attributes.

For that reason, I work within an Outcome-Based Education (OBE) structure in which each course

²Washington Accord is an international agreement among accreditation bodies that recognizes the substantial equivalence of accredited engineering degree programs and the graduate attributes expected of professional engineers.

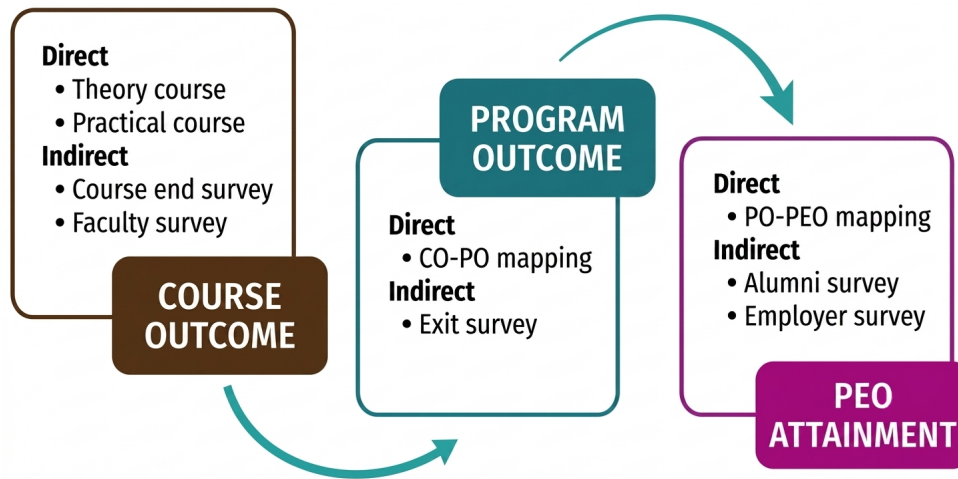


Figure 1: Framework for Mapping CO, PO and PEO in OBE.

contributes to the broader graduate profile (Noushad 2024; Mahbubul Syeed et al. 2022). The graduate profile defines the capabilities students should demonstrate by the time they complete the program. From there, I develop Course Outcomes (COs), align them with Program Outcomes (POs), and design assessments that generate evidence of learning rather than simple task completion. This alignment helps me keep the course coherent from start to finish (see Appendix A).

In my courses, this structure begins with Program Educational Objectives (PEOs), which describe what graduates should become within a few years of graduation. Program Outcomes translate those longer-term aims into measurable attributes, and Course Outcomes specify how an individual course contributes to them. I write COs using Bloom's taxonomy so that the level of cognitive demand is explicit. This gives my assessment design a clear logic and ensures that students are building knowledge, skills, and professional judgment in a deliberate sequence (Sunra, Aeni, and Sally 2024; Katawazai 2021).

Figure 1 shows how I use this framework in practice. Every assessment has a defined purpose within the OBE structure. I begin at the course level by measuring COs through direct methods such as class tests, assignments, labs, projects, and presentations, and through indirect methods such as student feedback. In my Digital Signal Processing course, for example, CO1 and CO2 are measured through algorithm implementation tasks, while student surveys help me judge clarity, relevance, and perceived difficulty.

These course outcomes are then mapped to Program Outcomes (see Appendix B). When students design and implement signal-processing algorithms, they are not only completing a task. They are demonstrating broader competencies such as engineering knowledge and problem analysis. I also use feedback from exit surveys to judge whether students feel prepared to apply those skills beyond the classroom.

The alignment continues to the PEO level. PEOs describe the longer-term development expected of graduates, such as becoming effective problem solvers, competent professionals, and lifelong learners. Evidence at this level comes from alumni and employer feedback. When former students report using Digital Signal Processing concepts in professional projects, that strengthens the case that the course contributes to those broader educational objectives.

This structure affects how I choose assessment methods. I use Bloom's taxonomy to match the assessment type to the intended level of learning (Adams 2015; Thompson and Wheeler 2008). Lower-level outcomes such as remembering and understanding are checked through quizzes and short tests. Higher-level outcomes such as analysis, evaluation, and design are assessed through case studies, open-ended problem solving, simulation tasks, and projects. In circuit analysis, for instance, I often begin with short conceptual checks and then move toward simulation-based assignments that require students to apply, interpret, and evaluate.

Rubrics are central to this process. I use them not only to score work, but also to make the OBE structure visible to students. Each criterion in a rubric is tied to a specific outcome, so students can see what is being measured and why. This makes assessment more transparent and gives me a practical way to revise courses when program expectations change (Noushad 2024).

I use both formative and summative assessment throughout my courses. Formative assessment includes in-class problem solving, peer discussion, short reflections, coding tasks, debugging exercises, and quick quizzes. These help me identify misconceptions early and adjust teaching before gaps become entrenched. Summative assessment includes examinations, projects, labs, reports, and presentations. Together, these methods allow me to assess not only what students know, but also how they apply knowledge, explain decisions, and respond to technical uncertainty.

Continuous improvement is built into this process. I review attainment data against CO and PO targets after each term and use that evidence to adjust course design. When I found that students were struggling with Fourier–Laplace transitions, I introduced scaffolded tasks earlier in the semester. The following cohort showed measurable improvement on the same outcome. This cycle of identifying gaps, adjusting instruction, and reassessing the results keeps assessment responsive and useful (West 2023; Widiana et al. 2023).

Table 1 summarizes courses I have developed and taught, together with the teaching methods, assessment tools, and teaching level.

Table 1: Courses Taught at KIIT University, India (2009–2010)

Course	Time	Teaching methods	Assessment tools	Level
EC 504 Digital Signal Processing	2009–2010	Class lectures with presentation and whiteboard, problem-solving, case discussions, flipped classroom, simulation with MATLAB	Class tests, projects, presentations, coding assignments, design reviews, assignments, mid-term examinations, final examination	Bachelor
EC 602 VLSI Design	2009–2010	Class lectures with presentation and whiteboard, blended learning, case work, fabrication videos, packaging, flipped sessions, class projects	Class tests, class projects, homework, simulation exercises, assignments, mid-term examinations, final examination	Bachelor
EC 703 Computer and Communication Networks	2009–2010	Class lectures with presentations and whiteboard, problem-solving activities, case discussions, blended learning, flipped-classroom, consulting projects, company visits, hands-on sessions with networking devices	Class tests, class projects, quizzes, assignment portfolio, prototype demonstrations, simulation tasks, mid-term examinations, final examination	Bachelor

Table 2: Courses Taught at UITS, Bangladesh (2011–2022)

Course	Time	Teaching methods	Assessment tools	Level
ECE 429 Wireless Communication	2012–2022	Lectures with presentations and whiteboard, problem-solving activities, case discussions, blended learning, flipped-classroom tasks, consulting projects, company visits, hands-on work with networking devices	Class tests, projects, presentations, homework, assignments, mid-term examinations, final examination	Bachelor

Course	Time	Teaching methods	Assessment tools	Level
EEE 207 Electronic Measurement and Instrumentation	2015–2017	Lectures with demonstrations, lab experiments, instrumentation setup practice, case discussions, blended learning, hands-on measurement sessions	Lab reports, class tests, quizzes, assignments, viva, mid-term examinations, final examination	Bachelor
EEE 313 Microprocessor and Microcontrollers	2015–2017	Lectures with board work, coding demonstrations, lab programming, hardware interfacing, project-based learning, flipped-classroom tasks	Coding assignments, lab tests, quizzes, class tests, projects, viva, mid-term examinations, final examination	Bachelor
EEE 253 Computer Networking and Data Communication	2014–2018	Lectures with diagrams, case discussions, packet-tracing labs, network configuration exercises, blended learning, hands-on work with routers and switches	Lab reports, configuration tasks, quizzes, class tests, projects, mid-term examinations, final examination	Bachelor
EEE 309 Communication Theory	2011–2020	Lectures with mathematical derivations, simulation demonstrations, problem-solving activities, case discussions, blended learning	Quizzes, class tests, MATLAB assignments, presentations, mid-term examinations, final examination	Bachelor
EEE 315 Digital Signal Processing	2012–2020	Lectures with board explanations, MATLAB simulations, coding tasks, problem-solving sessions, flipped-classroom activities	MATLAB tasks, quizzes, class tests, mini-projects, mid-term examinations, final examination	Bachelor
EEE 305 Signal and Systems	2011–2018	Lectures with board work, problem-solving, MATLAB-based demonstrations, conceptual discussions	Quizzes, class tests, problem-solving assignments, mid-term examinations, final examination	Bachelor
EEE 431 Optical Fiber Communication	2014–2016	Lectures with diagrams, hardware demonstrations, simulation tasks, case discussions	Class tests, quizzes, assignments, project work, mid-term examinations, final examination	Bachelor
EEE 151 Electrical Engineering	2016–2018	Lectures with board work, problem-solving, demonstrations, basic lab activities	Quizzes, class tests, lab assessments, assignments, mid-term examinations, final examination	Bachelor
EEE 108 Circuit Simulation	2018–2020	Lectures with software demonstrations, simulation exercises, design tasks, hands-on circuit modeling	Simulation tasks, lab reports, quizzes, class tests, projects, mid-term examinations, final examination	Bachelor
ECE 544 Data Communication (new course introduced and developed)	2018–2022	Lectures with diagrams, case discussions, protocol analysis, simulation tasks	Quizzes, class tests, protocol analysis tasks, assignments, mid-term examinations, final examination	Master
ECE 540 Telecommunication Networks	2018–2022	Lectures with network design problems, case studies, simulation tasks, configuration exercises	Quizzes, class tests, simulation tasks, design assignments, mid-term examinations, final examination	Master
ECE 542 Advanced Telecommunication Networks	2018–2022	Lectures with advanced case discussions, simulation studies, protocol evaluations	Quizzes, class tests, simulation tasks, protocol evaluation assignments, mid-term examinations, final examination	Master
ECE 463 Satellite Communication	2018–2022	Lectures with link-budget problems, case discussions, simulation tasks	Quizzes, class tests, link-budget tasks, assignments, mid-term examinations, final examination	Master

Course	Time	Teaching methods	Assessment tools	Level
ECE 545 Digital Modulation Techniques	2018–2022	Lectures with mathematical analysis, simulation tasks, problem-solving activities	Quizzes, class tests, simulation tasks, assignments, mid-term examinations, final examination	Master

Table 3: Courses Taught at Suntec Sanyo Engineering (Japan) (2018–2019)

Course	Time	Teaching methods	Assessment tools	Level
Overhead Transmission Line Design and Construction	2018–2019	Class lectures with presentations and whiteboard, problem-solving activities, case discussions, consulting projects, company, industry, and site visits, hands-on sessions with high-voltage devices	Class quizzes, class tests, formative assessment through questioning, practical assessment, oral examinations	Engineers and professionals

In theory-heavy courses such as Signals and Systems (EEE 305), I begin assessment at the lower and middle levels of Bloom’s taxonomy because students first need conceptual clarity before they can solve more complex problems. Early quizzes and class tests help me diagnose misconceptions about convolution, system linearity, frequency response, or Laplace-domain behavior. I use those results to adjust lectures, examples, and homework while the course is still in progress. By the time students reach the final examination, the assessment focuses less on formula recall and more on interpretation, application, and analysis. That progression is intentional.

Table 4: EEE 315: Digital Signal Processing

CO No.	CO Statement	Domain	Delivery method	Assessment Tools
1.	Apply the principles of discrete-time signal analysis to perform various signal operations.	Cognitive/Apply	Class lectures	Class tests, homework, final examination
2.	Apply the principles of Fourier transform analysis to describe the frequency characteristics of discrete-time signals and systems.	Cognitive/Apply	Class lectures	Class tests, homework, final examination
3.	Use computer programming tools (usually MATLAB or C/C++) to process and visualize signals.	Cognitive/Apply, Affective/Receiving Phenomena	Class lectures	Class tests, homework, final examination

Laboratory courses require a different assessment logic. In Communication Theory Lab (EEE 312), the emphasis is not only on analytical understanding but also on experimental skill, interpretation of results, technical reporting, and oral explanation during viva examinations. A lab on amplitude modulation is not complete when the waveform appears on the oscilloscope. It is complete when students can explain sideband behavior, discuss distortion in the envelope detector, and compare measured SNR with theoretical predictions. I have found that this kind of assessment builds engineering intuition more effectively than written tests alone.

I follow a similar pattern in Digital Signal Processing. Students first work through transformations by hand, then implement them in MATLAB, and finally explain where simulation results diverge from ideal theory because of sampling, quantization, or windowing. Assessing this full sequence, concept, implementation, and interpretation, helps me evaluate how students think as well as what they know.

Table 5: EEE 316: Digital Signal Processing Lab

CO No.	CO Statement	Domain	Delivery method	Assessment Tools
1.	Illustrate the handling of continuous signals using MATLAB.	Cognitive/Understand	Class lectures and practical demonstration	Lab performance, quiz, viva, assignment, project, presentation
2.	Illustrate the handling of discrete/digital signals using MATLAB.	Cognitive/Understand	Class lectures and practical demonstration	Lab performance, quiz, viva, assignment, project, presentation

Table 6: EEE 490: Thesis/Project/Internship

CO No.	CO Statement	Domain	Delivery method	Assessment Tools
1.	Review research literature relevant to engineering problems.	Cognitive/Understand	Guided literature reviews, database search workshops, annotated bibliographies, and key paper discussions.	Teamwork, communication skills, interview, survey, observation records, monitoring of skill development, presentation
2.	Investigate complex problems using research-based knowledge and research methods.	Cognitive/Analyze	Research design workshops, problem-based learning, case studies, and supervised data collection.	Teamwork, communication skills, interviews, survey, observation records, monitoring of skill development, presentation
3.	Select and apply appropriate resources and/or modern engineering tools.	Cognitive/Apply	Simulation software training, lab demonstrations, project-based applications, and peer-learning.	Teamwork, communication skills, interviews, survey, observation records, monitoring of skill development, presentation
4.	Write effective reports and make effective presentations.	Cognitive/Remember, Affective/Responding to phenomena	Academic writing workshops, peer-review sessions, presentation rehearsals, and supervised report drafting.	Meeting deadlines, report writing, and team presentations
5.	Demonstrate the depth for continuous learning.	Cognitive/Apply, Affective/Valuing, Psychomotor/Complex Overt Response	Reflective journals, self-directed tasks, research seminar participation, and collaborative discussions on emerging technologies.	Teamwork, communication skill, interview, survey, observation records, monitoring of skill development, presentation

In design-intensive courses such as Microprocessor and Microcontroller Systems (EEE 313), assessment has to capture higher levels of Bloom's taxonomy, especially design, evaluation, and creation. Students build working systems, debug hardware, and defend design decisions during viva examinations. I use staged project milestones, code reviews, and iterative submissions so that I can assess progress as well as the final result. A team developing a temperature-controlled fan system, for example, may submit an initial design sketch, then an interrupt-driven prototype, and later an optimized version with improved timing, reliability, and documentation. When I ask why they selected a particular timer mode or ADC sampling interval, I am assessing design reasoning, not only the finished device.

Assessment in thesis and internship courses requires another layer of judgment. Here I focus on research design, methodological rigor, communication, and the ability to work through uncertainty. My assessment includes supervision logs, proposal defenses, progress presentations, and a final oral examination. If a student discovers midway through a project that a chosen sensor does not have sufficient dynamic range, I do not treat that only as an error. I assess how well the student diagnoses the problem, revises the design, and justifies the new direction. That is much closer to real engineering practice.

Across all courses, formative assessment plays a major role in my teaching decisions. Short analytical problems, in-class design tasks, debugging exercises, and reflective notes help me identify learning gaps while there is still time to respond. If several groups misunderstand noise behavior in a communication link, I redesign the next class around a targeted simulation and comparison task. Summative assessment then allows me to judge cumulative learning through midterms, finals, projects, labs, and presentations. Rubrics tie these judgments back to the OBE structure. Criteria such as accuracy, reasoning, professionalism, and design clarity are each linked to specific COs and POs. Students receive these rubrics in advance, so expectations are explicit and feedback is easier to interpret.

This alignment is also visible in how course data are used. Individual questions and tasks are mapped to specific COs, and CO attainment is then mapped to relevant POs (see Appendix A). When student performance is aggregated across these tasks, I can judge how well each outcome has been achieved and whether the course is contributing effectively to broader program goals. I use a threshold of 60% as the minimum level for satisfactory attainment. Scores above that point indicate that the intended competency has been demonstrated at an acceptable or stronger level.

What matters to me is not the percentage alone, but what the percentage represents. This cascading alignment, from assessment task to CO, from CO to PO, and from PO to PEO, ensures that every activity serves a purpose beyond grading. It turns assessment into evidence for curriculum quality and student development. It also helps me document whether a course is actually doing what it claims to do.

I use these results to improve the next iteration of the course. After each term, I review CO achievement patterns to identify weak areas. When students underperformed on Fourier–Laplace transitions in Signals and Systems, I added scaffolded formative exercises earlier in the semester. The next cohort performed better on that outcome. That kind of revision is routine in my teaching. Assessment, for me, is not a final judgment. It is a structured way to connect expectation, evidence, and improvement.

My teaching methods are built around this same logic. A safe and positive classroom allows students to engage with challenging engineering problems without withdrawing from the process. Active-learning tasks such as real-time problem solving, interactive demonstrations, coding exercises, and short quizzes help students apply concepts while they are still learning them. In Digital Signal Processing, for example, I ask students to implement algorithms on real signal data during class. This supports CO1 and CO2 while also strengthening broader competencies such as engineering knowledge and use of modern tools.

Collaborative learning extends that process. Group assignments and peer discussion require students to explain ideas, challenge assumptions, and solve design problems together. That strengthens technical understanding, but it also develops teamwork and communication. These are not secondary skills. They are central to engineering practice and directly support the broader educational objectives of the program. In that sense, my teaching methods and assessment methods are part of the same design. Both are meant to help students grow into capable, thoughtful, and professionally prepared engineers.

Pedagogical Materials

I design pedagogical materials around one expectation: engineering students learn best by doing. The resources I prepare are meant to move students from passive understanding to hands-on experimentation, digital problem-solving, and iterative design. My goal is to make abstract theory concrete enough to test, modify, and question.

In theory-heavy courses, I build materials that connect mathematics to physical behavior. When teaching convolution, Fourier–Laplace transitions, or sampling, I avoid long derivations without context. I begin with short conceptual diagnostics that reveal misconceptions early. Students then work with MATLAB notebooks, interactive scripts, and guided simulation environments where they change parameters and observe the effect immediately. In a lesson on aliasing, for example, students vary the sampling rate and watch folding frequencies appear in the spectrum. In filter design and windowing tasks, they use editable MATLAB templates that allow them to add noise, distort signals, and compare theoretical predictions with actual outputs. These materials make theory easier to test and harder to treat as something purely symbolic.

I also prepare Python-based notebooks for students who prefer open-source tools. In Signal Processing, DSP, and Communication Theory, these notebooks include visualization tools that let students sweep frequencies, change filter order, and introduce quantization effects interactively. I usually leave parts of the notebook incomplete so students have to finish the code, interpret the results, and justify their design choices. That mirrors real engineering work, where information is often incomplete and judgment matters.

In hardware-based courses, my materials focus on physical experimentation. In Microprocessor and Microcontroller Systems, I provide modular hardware templates, annotated circuit diagrams, and debugging guides. Students use these to build interrupt-driven systems, communication routines, and sensor interfaces. In one material package, for example, students connect a temperature sensor, process the signal through an ADC, and visualize the result on a MATLAB or Python dashboard. I also require them to keep troubleshooting logs that document each failure, the likely cause, and the corrective step. That practice builds habits they will need in professional engineering work.

Networking and wireless communication courses require a different setup. For those courses, I build digital environments that simulate real infrastructure. I prepare virtual labs using Cisco Packet Tracer, GNS3, and cloud-based tools where students configure routers, build network topologies, analyze packet flow, diagnose routing loops, and measure congestion. I often place faulty configurations in the lab files so students have to locate and fix the problem themselves. I also use spectrum visualization tools and software-defined radio (SDR) simulations in which students model channel impairments and test modulation schemes using real I/Q data. These resources make wireless communication experimental even when full RF hardware is not available.

I also use short “micro-labs” inside regular lectures. These are 15–20-minute hands-on tasks that give students an immediate technical problem to solve. They might analyze a noisy communication link, debug an ADC routine, or design a simple FIR filter using an interactive interface. These tasks break up long lectures and bring students back into active thinking before attention drops. In some courses, I also use real datasets from sensor networks, IoT systems, or communication logs so students can work with practical data rather than ideal textbook examples. When teaching noise and interference, for instance, I provide recorded waveform data from an SDR-based setup and ask students to identify and quantify the disturbances.

My collaborative materials support the same hands-on approach. I prepare peer-review forms, design critique sheets, and structured problem-solving templates that help students explain assumptions, compare design choices, and evaluate trade-offs with teammates. In project-based courses, I also use Git repositories for code review, shared engineering notebooks, and online simulation dashboards so that teams can track decisions, failures, and design changes clearly. These tools help make the classroom feel closer to an actual engineering workspace.

Assessment materials are also aligned with OBE principles. My rubrics, lab templates, and project briefs are designed to guide student thinking, not just assign marks. Each criterion is linked to a CO and PO, so expectations are explicit from the start. In hands-on tasks, I assess diagnostic reasoning, interpretation of results, design clarity, and the ability to justify alternatives. In simulation-based work, I assess correctness, reproducibility, code quality, and the validity of the conclusions. Students can see how each task contributes to the broader competencies expected of graduate engineers.

These materials change from semester to semester. When I see persistent weaknesses, such as poor control-system reasoning or weak interpretation of network behavior, I revise the next version of the course materials to address those areas directly. That may mean earlier intervention tasks, stronger visualizations, or additional micro-labs. This process helps me improve conceptual understanding and generate clearer evidence of PO achievement across cohorts.

In the coming years, I plan to expand the digital and hands-on elements further by adding remote-access SDR laboratories, browser-based simulators, cloud-hosted microcontroller environments, and virtual communication networks. These additions would give students more flexibility to work with realistic engineering systems beyond scheduled class time.

Taken together, my pedagogical materials form a connected system: lab tasks, simulations, code templates, conceptual diagnostics, peer-review structures, and OBE-aligned assessment tools. Their purpose is simple. Students should not only learn engineering concepts. They should test them, apply them, question them, and use them like engineers.

Education Management

My first faculty appointment was as a Lecturer in 2010, and later that year I was promoted to Assistant Professor. The promotion came early and reflected the institution's confidence in my work. From the beginning, I was assigned final-year teaching responsibilities, which are not usually given to early-career faculty. In those classes, I saw a clear gap between theoretical understanding and practical application. Students could explain concepts, but many struggled to connect them to real engineering problems. I responded by redesigning parts of the courses. I introduced real-life examples, clearer learning sequences, hands-on projects, experiments, and early research activities. This shifted the courses toward application-driven learning, and students later identified these changes as examples of good practice within the department.

In 2011, I joined UITS as an Assistant Professor. Over more than a decade, I taught undergraduate and postgraduate students, developed laboratories, and supervised large numbers of engineering students. I taught courses for students from Electronics and Telecommunication Engineering, Electrical Engineering, Electronic and Electrical Engineering, Electronics and Communication Engineering, Computer Science and Engineering, and Information Technology. Teaching across these programs required breadth across electronics, electrical systems, communication, networking, software, programming, machine learning, and signal processing. It also required flexibility in teaching method, course design, and student support. That experience strengthened my ability to adapt to different academic levels, backgrounds, and disciplinary needs.

My work at UITS gradually expanded beyond teaching. The university began assigning me roles that required academic leadership and administrative responsibility. This did not happen suddenly. It developed over time through trust, consistency, and evidence that I could manage both academic and institutional tasks effectively.

I started by serving on a number of academic and administrative committees between 2011 and 2022. These roles gave me a solid understanding of institutional governance and collaborative decision-making. In 2017–2018, I was appointed to the Self-Assessment Committee, where I also provided pedagogical training to faculty members in the department. Later, I served as Member Secretary to the Faculty (2018–2022), Advisor to the IT Office (2018–2022), Member of the Finance Committee (2020–2022), and Editorial Board Member (2020–2022). These roles involved work on

academic quality, financial oversight, and institutional decision-making.

At the same time, I took on more operational leadership roles. I served as Project Manager for the university's Enterprise Resource Planning initiative (2020–2022), Director of the Information and Communication Technology Cell (2020–2022), and Head of the Department of EEE (2019–2022). These positions placed me at the center of curriculum reform, digital transformation, and the implementation of outcome-based education across the institution. During this period, I authored the university's OBE Handbook, redesigned assessment systems, updated program structures, and established several institutional units, including the Training and Placement Cell (2020) and the ICT Cell (2020). I also trained faculty members across multiple departments in teaching, assessment, and quality assurance practices. Through this work, I grew into a role that combined educational leadership with institutional development. The trust placed in me extended beyond academic administration. I also designed the official university logo, which reflected the broader confidence the institution had in my work.

As Director of the ICT Cell (2020–2022), I led a university-wide digital transformation aimed at improving quality, efficiency, and transparency in education delivery. One major outcome was the design and implementation of University Comprehensive Academic Management (UCAM), a unified platform for academic operations, student services, and research management. This included ERP functions for scheduling, performance tracking, and administrative coordination, along with centralized repositories for faculty and student research.

This work became especially important during the COVID-19 pandemic. When digital infrastructure became essential for continuity, I worked to standardize remote teaching platforms, expand access to digital tools, and provide technical and pedagogical support to faculty and students. These efforts helped maintain teaching continuity and also improved the quality of online learning through interactive modules, real-time analytics, and automated assessment tools.

Pedagogical Training and Curriculum Development

At UITS, my work in pedagogical training and curriculum development began with active participation in institutional workshops and later expanded into academic leadership and reform. I took part in a series of quality-assurance and teaching-development workshops organized through the Internal Quality Assurance Cell (IQAC), which supported the university's transition toward OBE (see Appendix D). These workshops introduced structured approaches to academic governance, self-assessment, collaborative decision-making, and continuous improvement. Sessions on team building, curriculum models, self-assessment processes, and teaching methods, led by senior academics from UGC, BUET, UIU, NSU, and other universities, sharpened my understanding of learner-centered course design and outcome-aligned assessment. This training strengthened my ability to combine theoretical rigor with practical competence and helped me apply active learning more systematically across the engineering curriculum.

As my involvement deepened, I moved from participant to facilitator. I conducted workshops on survey interpretation and stakeholder feedback analysis, helping faculty members and academic units use evidence to improve teaching and learning. I also developed and administered surveys for students, staff, and alumni as part of the self-assessment process, so that curriculum decisions could be based on data rather than assumption. This work improved my own pedagogical practice, but it also helped strengthen a broader culture of reflective teaching within the department.

This background in pedagogical training supported my later work as Head of the Department of Electronics and Communication Engineering, where I led major curriculum reform. Under my leadership, the Bachelor of Science in Electronics and Communication Engineering was comprehensively revised. I updated course structures, removed outdated content, and introduced new modules in IoT systems, advanced wireless communication, embedded systems, and related emerging areas. These changes were driven by a clear need: engineering graduates now require cross-disciplinary

knowledge, practical skill, and strong analytical ability if they are to work effectively in a changing technological environment.

I also initiated and designed the Master of Science in Telecommunication Engineering in response to growing national demand for expertise in telecommunications. Bangladesh's digital infrastructure was expanding rapidly, and the industry needed engineers who could work with advanced communication systems and next-generation networks. The program was designed to combine advanced theory with research-oriented study and included areas such as optical fiber communication, satellite systems, and 5G technologies. It created a stronger pathway from undergraduate foundations to specialized and professionally relevant expertise. Establishing this program increased both the academic depth and the strategic relevance of the department.

These experiences, from pedagogical workshops to curriculum reform and program development, have shaped how I see my role in higher education. I do not see teaching as limited to the classroom. I also see it as the work of building systems, structures, and academic environments that help students learn well and prepare them for meaningful professional contribution.

Evidence of Student Learning

I judge student learning by how students think, design, justify decisions, and improve over time. In engineering, that evidence rarely comes from one exam. It appears more clearly in how students handle laboratory work, interpret data, debug implementations, and explain their choices with increasing precision. My courses are built around clear learning outcomes, and each activity, lab, and assessment is aligned with those outcomes.

The evidence presented for EEE 315: Digital Signal Processing shows strong attainment of course outcomes and clear alignment with the graduate profile defined by the PEOs and POs (see Appendix B). The reported CO achievement, CO1 at 91%, CO2 at 100%, and CO3 at 87%, indicates that students can apply engineering knowledge, analyze DSP problems, and use modern tools effectively. These outcomes align with PO (a), PO (b), and PO (e), and they support PEO1 through demonstrated problem-solving ability. They also support PEO2 by showing that students are developing practical competence and technical confidence. The strong performance in simulation and iterative design work also supports PEO3 by encouraging adaptability and continued learning.

If the results had been weaker, the implications would have been clear. A low score in PO (e), for example, would indicate inadequate exposure to modern engineering tools and would require additional lab sessions or software-based exercises. A weak result in CO2 would suggest the need for more structured analytical tasks and guided problem-solving. In that sense, attainment data help me judge both immediate course performance and broader progress toward the program's educational objectives.

Even when a class reaches the expected CO, PO, and PEO thresholds, individual student results still matter. They often reveal where specific competencies are weaker (see Appendix B, Figure B5). In one case, a student achieved 53% in CO1, which was below the satisfactory benchmark, while doing better in CO2 (75%) and CO3 (58%). In another case, a student scored 58% in CO3 but achieved 75% in CO1 and 91% in CO2. These patterns show that strong class-level attainment can still mask individual difficulties in foundational knowledge or modern tool usage.

I expect this kind of variation in a course such as Digital Signal Processing because it demands both conceptual rigor and tool-based implementation. Students who struggle in CO1 often have trouble turning mathematical principles into problem-solving strategies, which affects PO (a): Engineering Knowledge. Students who struggle in CO3 usually have difficulty with simulation environments, implementation logic, or algorithmic design, which relates more directly to PO (e): Modern Tool Usage. If I ignore those gaps, they can affect progress toward PEO1 and PEO2.

To address that, I use targeted intervention. Students who struggle with CO1 receive short remedial sessions and guided problem-solving focused on mathematical fundamentals. Students with

weaker CO3 performance receive additional lab practice in MATLAB and Python, often supported by video demonstrations and peer-assisted work. I also use short follow-up assessments and individualized feedback before the next major evaluation. These interventions are meant to improve course-level attainment, but they also build resilience and adaptability, which support PEO3.

This reflects a broader principle in my teaching. Evidence of learning is not only an aggregate number. It is also the extent to which each student is moving closer to the graduate profile defined by the program. When students who initially struggled later show confidence in signal analysis or simulation-based implementation, that shift gives me stronger evidence of learning than a final score alone.

Because my courses emphasize complex problem-solving and applied reasoning, I design assessments that resemble real engineering work. This supports PO (a): Engineering Knowledge, PO (b): Problem Analysis, and PEO1: Problem Solving. Students usually work in iterations rather than submitting one final polished answer. In MATLAB-based DSP tasks and SDR simulations, for example, they begin with incomplete or procedural implementations. I give immediate technical feedback, and they revise, retest, and justify their changes. By the end, they are not simply running code. They are identifying spectral artifacts, improving signal paths, and refining interrupt routines with growing confidence. That progression tells me much more than a one-time exam because it reflects how engineering problems are actually solved.

Reflection is another source of evidence. After major labs, I ask students to write short analytical notes explaining not what they did, but why the system behaved as it did. These reflections often reveal a level of understanding that closed questions do not. In DSP, students move from plotting spectra to explaining leakage, aliasing, and filter behavior with proper justification. In communication courses, they begin to discuss channel impairments, quantization effects, and link-budget decisions more accurately. That movement from description to explanation is one of the clearest signs that both learning outcomes and graduate attributes are being reached.

Mid-semester and end-semester evaluations provide another layer of evidence, especially for PO (j): Communication and PO (i): Individual Work and Teamwork. I use open-ended prompts because students often explain their learning more clearly when they can describe where they struggled, what helped, and which activities made a difference. Questions such as “*What activity most improved your understanding?*” and “*What part of the course should be redesigned?*” give me feedback that I can act on directly. When I teach the same course again, I show students what was changed based on earlier feedback. That practice builds trust and also models continuous improvement, which supports PEO2.

External evaluators provide another useful perspective. In courses with practical labs or communication-system projects, I discuss student performance with external examiners. Their comments on design choices, coding practices, and modeling assumptions often confirm the progress I see in class. These discussions help me judge whether students are meeting professional expectations and whether the course is preparing them well for industry.

Some of the strongest evidence comes after graduation. Former students often tell me that the MATLAB workflows in DSP, the debugging habits developed in microcontroller labs, or the practical routing work in networking courses helped them during internships and early employment. That feedback matters because it shows that course outcomes are carrying into professional settings. It also supports the broader goal of preparing graduates who continue learning after formal study.

I also see evidence of learning in the classroom itself. At the start of a semester, many students hesitate when they face open-ended engineering problems. By the end, they approach the same kind of problem more systematically. They form hypotheses, break systems into parts, interpret anomalies, and test alternatives with more confidence. That change in behavior is one of the clearest signs that learning has taken place.

Supervision

I have explored and applied several approaches to become an effective supervisor for bachelor and master projects and theses (Lee, 2012; McCarthy, 2015; Pearson & Brew, 2002). I do not see supervision as helping a student finish a document. I see it as mentoring, building confidence, and helping students become more independent thinkers. Over time, I have learned that good supervision requires structure, flexibility, and real attention to the student's development. It is also a collaborative process. Students learn from guidance, and I learn from how they question, interpret, and work through problems.

During my academic career, I supervised more than 200 bachelor students and over 50 master's theses at UITS between 2009 and 2022. A complete list is given in (see Appendix C). In many cases, I supervised jointly with colleagues, and I often encouraged students to work in teams where that format strengthened the project. Several bachelor and master students also worked with me on research activities that later contributed to published articles. Those collaborations were especially valuable because they turned supervision into a shared process of inquiry rather than a one-way transfer of advice.

My supervision style is structured, but not rigid. At the beginning, I set clear expectations about meetings, feedback timelines, and milestones. For master's students, I usually schedule more frequent meetings in the early stage so that the topic, research questions, and methodology are defined carefully. Because many of the theses I have supervised involved qualitative methods, I also set aside time after data collection to help students interpret the material and make sense of their findings. As the work progresses, I ask students to share drafts for feedback in person or online. I maintain an open-door attitude, but I am also clear about availability and scheduling. That clarity helps build trust and keeps the process manageable.

Because my teaching follows an outcome-based curriculum, my supervision practice is also tied to intended learning outcomes. At the start of a project, I focus on helping students review the literature effectively. I guide them through database searches and structured reading, often using IEEE Xplore or Scopus. I remember one bachelor student who struggled to narrow a topic until we worked through annotated bibliographies and discussed a small set of key papers together. The process helped the student gain both focus and confidence. As projects develop, I put more emphasis on problem-solving and research design. In one master's thesis on renewable energy systems, I encouraged the student to analyze real case studies before fixing the methodology. That improved the analytical strength of the work and supported the intended outcome of investigating complex problems through research-based knowledge.

I also treat the use of modern tools as an important supervisory outcome. In one project on power electronics, students first modeled circuits in simulation software before moving to laboratory implementation. That gave them a better grasp of system behavior and reduced trial-and-error in hardware. Communication is another part of supervision that I take seriously. I use academic writing support, draft review, and peer feedback to help students improve how they present technical work. In one case, I organized a mock presentation for a group of bachelor students and asked their peers to comment on clarity, structure, and technical depth. Their delivery improved noticeably after that session. I also encourage continuous learning by asking students to keep reflective notes and attend seminars when relevant. One master's student attended an international webinar on smart grids and later used those ideas to strengthen the thesis discussion.

Throughout the supervision process, assessment is continuous and aligned with OBE principles. I monitor progress, teamwork, communication, and independence through observation, discussion, and regular feedback. I do not treat these as formalities. They help students understand that completing the thesis is only one part of the process. Developing professional habits, research judgment, and self-direction matters just as much.

My supervision approach combines guidance with gradual independence. I also try to encultur-

ate students into research and professional communities by involving them in seminars, research discussions, and, when possible, co-authorship. This helps them see themselves not only as students completing a degree requirement, but as emerging members of a wider academic and engineering community.

Overall, I see supervision as a partnership. The aim is to help students produce strong academic work while also developing confidence, professional skill, and the habit of continuous learning. Every student is different, so I adapt my approach to fit the project and the person, while keeping the structure clear enough to support steady progress.

Continuous Improvement Through Students' Feedback

Student feedback is a central part of how I improve my teaching within the Outcome-Based Education framework. It helps me turn outcome mapping and assessment data into concrete changes in course design, teaching method, and learning resources. Comments from students across different courses have repeatedly pointed out where delivery was effective, where resources needed revision, and where assessment could be made clearer.

One recurring theme was the need for more real-life examples and practical exercises alongside theory. Students appreciated the structured weekly outlines and the balance between theory and application, but they also noted that some topics felt rushed and some parts of the syllabus were too dense. In response, I re-sequenced content to distribute the workload more evenly and added more industry-based case studies. In Digital Signal Processing, for example, I replaced generic examples with applications such as audio noise cancellation and biomedical signal analysis. That made concepts such as Fourier transforms easier to connect to practice. I also introduced short design challenges during class and optional formative quizzes so students could check understanding without the pressure of high-stakes grading.

Feedback on CO achievement showed that students understood the OBE structure but wanted to see the links between COs and assessments more clearly. I responded by adding a “*CO of the Week*” focus at the start of each module, where I linked the week’s topic to its relevant COs and POs. I also redesigned assignment and test cover sheets so that students could see which COs were being assessed in each question and how marks were distributed. Mid-semester CO progress reports gave students more individualized feedback and helped them see where improvement was needed.

Comments on learning resources led me to update both content and accessibility. I replaced outdated materials with more current sources, developed interactive MATLAB and Python tutorials, fixed broken links, and added a set of short video demonstrations for difficult topics. I also introduced visual aids and annotated solutions for practice problems. These changes made it easier for students to move between concept and application.

Feedback on teaching methods often focused on pacing and interactivity. Students wanted more group work and more regular checks for understanding. In response, I expanded the use of flipped-classroom elements and structured peer-learning activities. In Communication Theory, for example, students worked together on link-budget design tasks, which improved both teamwork and conceptual engagement. I also began ending classes with a short summary and a few diagnostic questions so I could judge whether the pace was appropriate.

Assessment was generally viewed as fair, but students wanted clearer rubrics and more opportunities to practice before major evaluations. I therefore began publishing detailed rubrics before major assignments, showing the criteria tied to each CO and the expected level of performance. Revision sessions before exams also changed. Instead of reviewing material broadly, I focused more on problem-solving strategy, which reduced student anxiety and improved confidence. I also created a set of self-paced exercises with annotated solutions so students could practice independently and receive immediate feedback.

These changes came directly from student feedback, and they reflect how I treat feedback in

general. I do not treat it as criticism to defend against. I treat it as evidence. Used properly, it improves the clarity, transparency, and effectiveness of the course. It also helps me keep teaching aligned with both student needs and program outcomes.

Dissemination

I see dissemination as part of academic responsibility. If a teaching practice works well, it should be shared, tested, and improved beyond one classroom. For that reason, I treat dissemination as a natural extension of reflective teaching rather than a separate activity.

One of the main ways I have done this is through institutional leadership and faculty development. As a member of the Self-Assessment Committee, a contributor to the university's OBE framework, and the author of the university's OBE Handbook, I trained faculty members across several engineering departments. These sessions focused on practical issues such as authentic assessment design and outcome mapping. My aim was not to present theory in the abstract, but to translate the teaching and assessment methods I use into steps that colleagues could adapt in their own courses.

I have also shared specific course practices through academic and professional development seminars. In sessions on active learning in engineering laboratories, for example, I showed how I redesigned Microprocessor course labs into problem-based formats and shared the project briefs, lab structures, and assessment rubrics that supported that change. I prefer sharing concrete materials and examples because they are easier for colleagues to evaluate and use than general claims about innovation.

My dissemination work also connects with research and industry engagement. Collaborative class projects have often produced useful case material for wider discussion. One example was an IoT-based energy monitoring project developed by mixed teams of younger students and senior diploma engineers. I later presented this model at a national engineering education conference to show how collaborative course design can support technical learning, peer mentoring, and industry connection at the same time. Work like this reflects a central belief in my teaching: engineering education should respond to real professional conditions.

Looking ahead, I want to extend this work further. I am developing a digital repository of pedagogical materials, including interactive MATLAB Live Scripts and simulation-based challenges, that can be shared as open educational resources. I also plan to collaborate with colleagues from NTNU and VUT on comparative work related to active learning in different educational and cultural contexts, including Bangladesh and Norway. I see this as part of a longer-term contribution to the scholarship of teaching and learning in engineering education.

Reflections on Own Educational Development

My path into academia began in 2009 at KIIT University, where I worked as a graduate teaching assistant and supported students in projects, research tasks, and thesis work. In those early experiences, I found that the most rewarding moments were not my own technical results, but the moments when students began to see a difficult idea clearly for themselves. That experience stayed with me. Guest lectures later that year, along with strong student evaluations, confirmed that teaching was not simply something I could do. It was the kind of work I wanted to keep doing.

As I gained experience, I realized that technical clarity and subject knowledge were not enough on their own. I could explain equations and circuit diagrams accurately, yet many students still struggled to turn those ideas into functioning systems or sound engineering reasoning. That was an important lesson. Knowledge does not automatically become competence. Students learn more deeply when they have to build, test, revise, and face the consequences of their own design choices. That realization marked my move away from lecture-centered teaching toward designing learning experiences.

One important stage in that development came from observing how students handled signal-processing tasks. Many could manipulate equations correctly, but they could not explain the spectral behavior they plotted or connect a formula to physical meaning. That pushed me to rethink how I taught these topics. By introducing MATLAB notebooks, SDR data, microcontroller tasks, and network simulators, I made invisible system behavior more visible and easier to investigate. Students began approaching aliasing, noise, filter behavior, and routing logic with more curiosity and less dependence on memorization. That experience reinforced a principle that still guides my teaching: engineering understanding becomes stronger when students control the tools and see the consequences of their own decisions.

Another turning point came when I began using conceptual diagnostics more deliberately. Instead of assuming that clear explanation meant clear understanding, I started designing classes with checkpoints that exposed misconceptions early. This changed how I sequence content, how I scaffold difficult topics, and how I adapt instruction while a course is still in progress. My teaching became more responsive because I could see where understanding was fragile before those weaknesses became fixed.

Feedback has also played a major role in my development. Mid-semester evaluations, open-ended student comments, and discussions with external examiners often reveal patterns that I would not otherwise see. Students pointed out where pacing worked, where labs needed better structure, and where some concepts needed stronger context. I began treating those responses like engineering data. I analyzed them, looked for repeated signals, and made changes accordingly. That turned course development into an iterative design process and improved my materials, communication, and assessment structure over time.

The growing role of digital tools also changed how I think about teaching. Over time, I moved from using isolated software tools to designing more integrated learning environments in which MATLAB, Python, Packet Tracer, GNS3, and microcontroller platforms support one another. That shift changed my role as a teacher. I was no longer only delivering content. I was building learning systems in which theory, simulation, and hardware practice could reinforce each other.

Collaborative learning deepened my understanding further. Watching students debate filter structures, troubleshoot implementation problems, and reason through communication-system behavior showed me how much learning happens through structured interaction. That led me to strengthen peer review, team-based analysis, and reflective tasks so that students had to justify choices rather than simply report answers.

Some of the most meaningful feedback has come years later from former students. Many have told me that the hands-on, tool-based, and diagnostic approach in my courses helped them deal with uncertainty during internships and early professional work. Those comments matter because they confirm that the habits developed in class, technical independence, iterative problem-solving, and engineering judgment, stay useful after graduation.

My development as an educator is still ongoing, but its direction is clearer now. I continue to work toward stronger alignment with learning outcomes, better digital learning environments, more authentic engineering tasks, and a more disciplined use of feedback. With each revision, I move closer to the kind of teacher I want to be, someone who helps students learn to think, design, test, and reason like engineers before they formally enter the profession.

Growing up surrounded by professors, I witnessed firsthand how busy their lives can be. I realized that most students never truly understand how busy professors are. The time they see professors in class is probably less than ten percent of the actual workload. At research universities, teaching a class is only the beginning. Professors prepare lectures, design assignments and exams, grade countless papers (a task that is far more time-consuming than most imagine), hold office hours, and, of course, show up to teach. Beyond that, they write grant proposals, review other people's work, publish their own research, supervise graduate students, and serve on evaluation committees, which often means reading hundreds of pages of dissertations for master's and PhD examinations. They

must constantly generate new research ideas, attend conferences, and still try to maintain a personal and family life.

Knowing all this, I often wondered *why anyone would choose to enter and stay in academia*. After seventeen years in academia, I realize that what made me stay is something deeper: the joy of discovery, the thrill of asking questions no one has answered before, and the privilege of finding solutions that can change lives. I stay for the moment when a student's eyes light up during office hours, when they arrive at the right answer, and when they finally understand *why* it is the right answer. I also stay for the quiet satisfaction of mentoring students and watching them grow into independent thinkers, and for the sense of contributing to knowledge that will outlast us. For me, that makes every challenge worthwhile.

* * *

Appendix A

Program Educational Objectives (PEO)

Program Educational Objectives (PEOs) are broad, forward-looking statements that define the professional achievements and career growth expected of graduates within a few years of completing the program.

- **PEO1, Problem Solving:** Ability to analyze complex engineering problems and propose effective solutions using appropriate analysis, synthesis, and engineering development tools.
- **PEO2, Competent Graduates:** Graduates will be innovative, competent, and productive in addressing industrial needs at home and abroad.
- **PEO3, Continuous Development:** Graduates will pursue lifelong learning and apply their knowledge responsibly in engineering practice and societal service, guided by strong moral and ethical values.

Program Outcome (PO)

Program Outcomes (POs) define the knowledge, skills, and professional behaviors students should demonstrate upon completing the program. These broad competencies, such as problem-solving, teamwork, and lifelong learning, are achieved through specific Course Outcomes (COs) that guide individual courses.

- **PO (a), Engineering Knowledge:** Ability to apply knowledge of mathematics, science, engineering fundamentals, and specialized engineering principles to solve complex engineering problems.
- **PO (b), Problem Analysis:** Ability to identify, formulate, review relevant literature, and analyze complex engineering problems in order to reach well-supported conclusions using the principles of mathematics, natural sciences, and engineering sciences.
- **PO (c), Design and Development of Solutions:** Ability to design solutions for complex engineering problems and to develop systems, components, or processes that meet specified needs with due consideration for public health and safety, as well as cultural, societal, and environmental factors.
- **PO (d), Investigation:** Ability to investigate complex engineering problems by identifying requirements, gathering relevant information, designing appropriate studies, and interpreting data to reach valid conclusions.
- **PO (e), Modern Tool Usage:** Ability to apply mathematical foundations, simulation methods, algorithmic principles, and engineering knowledge to model and design engineering systems using appropriate analysis, synthesis, and development tools, with an understanding of the trade-offs involved in design choices.
- **PO (f), The Engineer and Society:** Ability to understand human behavior, culture, and society through the study of humanities and social sciences, and to recognize the local and global impacts of engineering solutions.
- **PO (g), Environment and Sustainability:** Ability to understand the impact of professional engineering solutions on the environment and society, and to support sustainable development.

- **PO (h), Ethics:** Ability to apply ethical principles and commit to professional ethics, responsibilities, and the norms of engineering practice.
- **PO (i), Individual Work and Teamwork:** Ability to work effectively both as an individual and as a member or leader of a team in multifaceted and multidisciplinary settings.
- **PO (j), Communication:** Ability to communicate complex engineering ideas clearly and effectively in both speech and writing, including the preparation of reports and design documents, the delivery of presentations, and the exchange of clear instructions.
- **PO (k), Project Management and Finance:** Ability to demonstrate management skills and apply engineering principles in individual and team-based work, including the planning and management of projects in multidisciplinary environments.
- **PO (l), Lifelong Learning:** Ability to think critically and independently, exercise informed judgment in developing innovative engineering solutions, and recognize the need for continuous, self-directed professional development.

Table A1: Mapping of Program Outcomes to Program Educational Objectives

PO No.	Program Outcomes (PO)	PEO1	PEO2	PEO3
a.	Engineering Knowledge	yes	yes	
b.	Problem Analysis	yes	yes	
c.	Design/development of Solutions	yes	yes	
d.	Investigation	yes	yes	
e.	Modern Tool Usage	yes	yes	
f.	The Engineer and Society	yes		yes
g.	Environment and Sustainability	yes	yes	yes
h.	Ethics		yes	yes
i.	Individual Work and Teamwork		yes	yes
j.	Communication		yes	yes
k.	Project Management and Finance	yes		yes
l.	Lifelong Learning		yes	yes

Appendix B

Course Outcomes (CO)

Course Outcomes (COs) define what students should know, be able to do, and demonstrate by the end of a course. These outcomes are specific, measurable, and guide both teaching and assessment. For engineering students, COs typically combine technical skills, problem-solving abilities, and professional competencies, such as applying scientific and mathematical principles to solve complex problems, designing solutions, and conducting investigations using modern tools. They also include essential professional skills like effective communication, teamwork, ethical responsibility, and a commitment to lifelong learning. For example, following shows

Course Title	Digital Signal Processing
Course No	EEE 315

Course Outcome (CO)	CO Statement
CO1	Apply the principles of discrete-time signal analysis to perform various signal operations
CO2	Apply the principles of Fourier transform analysis to describe the frequency characteristics of discrete-time signals and systems.
CO3	Use computer programming tools (usually MATLAB or C/C++) to process and visualize signals

Program Outcome (PO)	PO Title	Corresponding CO's				
PO (a)	Engineering Knowledge	CO1	CO3			
PO (b)	Problem Analysis	CO2				
PO (c)	Design/Development of solutions					
PO (d)	Investigation					
PO (e)	Modern Tool Usage	CO2	CO3			
PO (f)	The Engineer and Society					
PO (g)	Environment and Sustainability					
PO (h)	Ethics					
PO (i)	Individual Work and Teamwork					
PO (j)	Communication					
PO (k)	Project Management and Finance					
PO (l)	Lifelong Learning					

Figure B1: Course Outcomes for EEE 315 and Their Alignment with Program Outcomes.

Class Test-1						
Question No.			Q1	Q2	Q3	
Select CO from Drop-down list -->			CO1	CO2	N/A	Total
Full Marks			15	10		25
Serial	Student ID	Name	Obtained Marks			
1	1010207000	Chandramohan Jayaram	9	8		17
2	1010207001	Deependran	12	7		19
3	1010207002	Ajay	11	6		17
4	1010207003	Ajay	10	9		19
5	1010207004	Abhinav	6	10		16
6	1010207005	Aravind	5	10		15
7	1010207006	Aravind	7	14		21
8	1010207007	Aravind	10	6		16
9	1010207008	Aravind	10	14		24
10	1010207009	Aravind	10	8		18
11	1010207010	Aravind	6	14		20
12	1010207011	Aravind	6	12		18
13	1010207012	Aravind	13	13		26
14	1010207013	Aravind	11	10		21
15	1010207014	Aravind	12	12		24
16	1010207015	Aravind	12	14		26
17	1010207016	Aravind	6	6		12
18	1010207017	Aravind	8	7		15
19	1010207018	Aravind	7	14		21
20	1010207019	Aravind	14	5		19
21	1010207020	Aravind	11	5		16
22	1010207021	Aravind	15	6		21
23	1010207022	Aravind	12	7		19
24	1010207023	Aravind	11	5		16
25	1010207024	Aravind	15	8		23
26	1010207025	Aravind	6	14		20
27	1010207026	Aravind	11	7		18
28	1010207027	Aravind	12	8		20
29	1010207028	Aravind	8	14		22
30	1010207029	Aravind	8	10		18
31	1010207030	Aravind	13	8		21
32	1010207031	Aravind	7	7		14
33	1010207032	Aravind	10	14		24
34	1010207033	Aravind	9	5		14
35	1010207034	Aravind	13	13		26
36	1010207035	Aravind	9	6		15
37	1010207036	Aravind	11	12		23
38	1010207037	Aravind	9	9		18
39	1010207038	Aravind	12	9		21
40	1010207039	Aravind	10	6		16
41	1010207040	Aravind	14	13		27
42	1010207041	Aravind	6	9		15
43	1010207042	Aravind	5	7		12
44	1010207043	Aravind	8	14		22
45	1010207044	Aravind	6	10		16

Figure B2: Evidence of Course Outcome Attainment in EEE 315 Based on Class Test Results

Assignment						
Select CO from Drop-down list -->			CO1	CO2	CO3	Total
Full Marks			5	5	5	15
Serial	Student ID	Name	Obtained Marks			
1	1019291010	G...	2	5	3	10
2	1019291011	D...	3	4	2	9
3	1019291012	...	2	2	5	9
4	1019291013	A...	4	4	5	13
5	1019291014	A...	5	3	2	10
6	1019291015	V...	5	2	2	9
7	1019291016	...	3	5	5	13
8	1019291017	...	4	5	2	11
9	1019291018	...	3	5	4	12
10	1019291019	A...	4	2	4	10
11	1019291020	...	4	2	2	8
12	1019291021	...	2	3	2	7
13	1019291022	...	4	3	4	11
14	1019291023	...	4	4	3	11
15	1019291024	...	3	2	3	8
16	1019291025	...	5	2	3	10
17	1019291026	...	5	4	2	11
18	1019291027	...	5	3	2	10
19	1019291028	...	3	3	4	10
20	1019291029	Y...	5	3	2	10
21	1019291030	...	5	2	4	11
22	1019291031	...	5	3	2	10
23	1019291032	...	2	4	4	10
24	1019291033	...	2	3	5	10
25	1019291034	...	5	5	2	12
26	1019291035	...	2	5	5	12
27	1019291036	...	3	4	2	9
28	1019291037	...	4	5	2	11
29	1019291038	...	3	2	4	9
30	1019291039	...	3	2	4	9
31	1019291040	...	5	5	3	13
32	1019291041	...	4	3	2	9
33	1019291042	...	2	2	4	8
34	1019291043	...	3	4	3	10
35	1019291044	...	3	3	4	10
36	1019291045	...	2	5	4	11
37	1019291046	...	2	5	3	10
38	1019291047	...	3	2	4	9
39	1019291048	...	2	2	3	7
40	1019291049	...	4	5	3	12
41	1019291050	...	5	4	2	11
42	1019291051	...	4	4	3	11
43	1019291052	...	3	3	4	10
44	1019291053	...	3	4	4	11
45	1019291054	...	5	3	5	13

Figure B3: Attainment of Course Outcomes in EEE 315 Through Assignment Performance

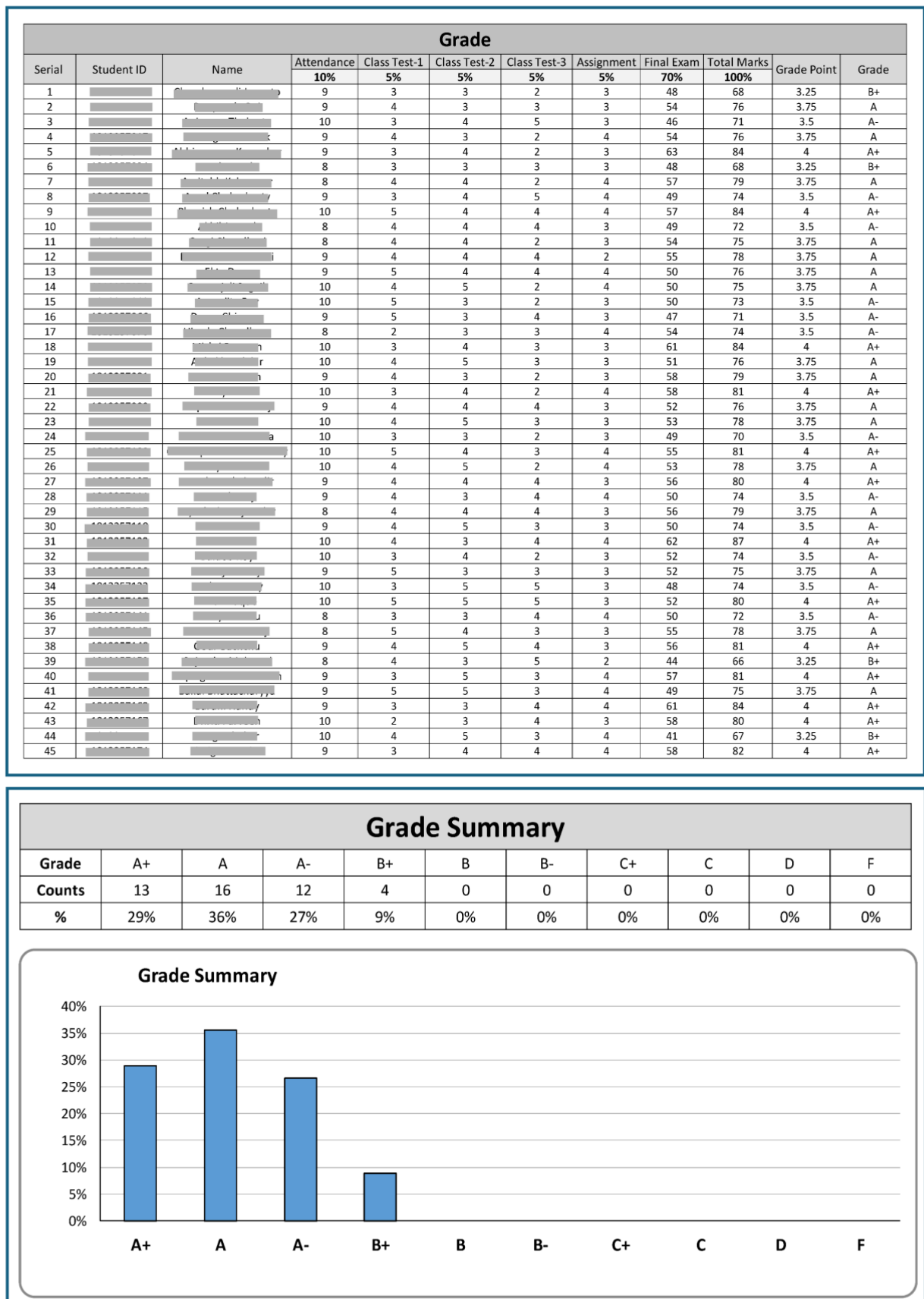


Figure B4: Evidence of CO Achievement for EEE 315 Based on Assignment Results

Individual CO Achievement								
Serial	Student ID	Name	CO1		CO2		CO3	
			Marks	Achievement	Marks	Achievement	Marks	Achievement
1			53%	0	75%	1	58%	0
2			78%	1	72%	1	65%	1
3			58%	0	62%	1	95%	1
4			68%	1	72%	1	68%	1
5			70%	1	86%	1	65%	1
6			65%	1	60%	1	65%	1
7			75%	1	91%	1	58%	0
8			65%	1	77%	1	75%	1
9			73%	1	92%	1	80%	1
10			68%	1	75%	1	73%	1
11			68%	1	85%	1	53%	0
12			60%	1	91%	1	63%	1
13			83%	1	80%	1	70%	1
14			68%	1	86%	1	60%	1
15			70%	1	74%	1	63%	1
16			70%	1	77%	1	78%	1
17			68%	1	72%	1	60%	1
18			73%	1	78%	1	73%	1
19			60%	1	89%	1	75%	1
20			95%	1	72%	1	50%	0
21			80%	1	74%	1	70%	1
22			85%	1	75%	1	73%	1
23							75%	1
24							65%	1
25							75%	1
26							58%	0
27							78%	1
28							73%	1
29							80%	1
30							78%	1
31							85%	1
32							60%	1
33							78%	1
34							73%	1
35							78%	1
36							75%	1
37							75%	1
38							78%	1
39							78%	1
40							70%	1
41							73%	1
42							88%	1
43							83%	1
44							58%	0
45							75%	1

CO Achievement Summary		
CO	CO Achievement	CO Achievement
CO1	91%	Y
CO2	100%	Y
CO3	87%	Y

PO Achievement Summary		
PO	PO Achievement	PO Achievement
PO (a)	93%	Y
PO (b)	100%	Y
PO (c)		
PO (d)		
PO (e)	87%	Y
PO (f)		
PO (g)		
PO (h)		
PO (i)		
PO (j)		
PO (k)		
PO (l)		

Figure B5: Overall PO Attainment for EEE 315 Based on CO Performance Metrics.

Appendix C

Student Supervision

During my tenure at UITS, I actively supervised both undergraduate and postgraduate students in their research, project work, and industrial internships. These projects span advanced topics in wireless communication, antenna design, signal processing, and emerging technologies such as 5G and IoT. My role involved guiding students through problem formulation, design, simulation, and implementation, while fostering critical thinking and professional competencies. The following list represents the students I mentored, and the research projects completed under my supervision.

Postgraduate Students

1. Md. Sohanur Rahman Adnan, Ishak Md. Tanjinul Haque, “*Reduction of Peak to Average Power Ratio in 5G Communication*”, (2021)
2. Md Golam Kibria, Md. Sabbir Hossain, Md. Kaisar Ahmad, “*Design & Implementation of Microstrip Patch Antenna for 5G Communication*”, (2021)
3. Abdulkadir Mohamed Mohamoud, Hassan Ahmed Mohamed, “*Design an H-shape microstrip patch antenna at dual band mm-wave for 5G communication*”, (2021)
4. Abdikani Mohamud Samatar Hussein, Abdinajib Jama Said, “*Interference Mitigation in Multiuser Multiple-Input-Multiple-Output Term-Evolution Advanced Systems*”, (2021)
5. Fath Ahmed Mohammed Ali, “*Design and Analysis of a Highly Sensitive Hollow Core Photonic Crystal Fiber for Chemical Sensing*”, (2021)
6. Abdulkadir Mumin Harare, Khadar Awil Abdi, “*Phased Array Antenna Design for 5G Network*”, (2021)
7. Mubarak Ali Hayd, “*Performance Analysis of Micro-strip Patch Antenna on Human Body at ISM(2.45GHz) Band*”, (2020)
8. Abdiaziz Mohamed, “*Performance Simulation and performance analysis of high-speed fiber optic communication system with OFDM/WDM*”, (2020)
9. Shankar Kumar Barman, “*Impact of Regulatory Aspects on 5G Mobile Communication Systems*”, (2020)
10. Sharmaarke Mohamed Ali, “*Performance Evaluation of LTE Physical Layer Using SC-FDMA & OFDMA*”, (2020)
11. Karar Jahidul Islam Anik, Riazul Ahsan Rabbi, “*Design and Implementation of Polyphase Filtering for 5G Communication using FPGA*”, (2020)
12. Farzana Hoque Sonia, Md. Matiwr Rahman, Md. Mydul Islam, “*Multiband Microstrip Antena Design for Wireless Energy Harvesting*”, (2019)
13. Sadia Shubash Mim, Mohammad Nurul Ambia, Baby Akhter, “*Distributed Energy Efficient Clustering (DEEC) for Wireless sensor networks*”, (2019)
14. Khadija Akther Sonya, Nadim Mohammad, “*How 5G technologies empower the health based on Internet of things*”, (2019)
15. Deen Mohammad, Md. Shamsul Arafin Shakil, “*Radar System in Vessel Traffic Management Information System*”, (2019)
16. Syeda Umme Honey, Md. Enamul Hasan, Sadman Islam, “*Design and Analysis of Massive MIMO for 5G Communication*”, (2019)
17. Md. Amanul Haque Aman, Md. Ahmadul Haque, “*Design and Implementation of a Secure Campus Network by using Different Security Mechanism and Routing Protocols*”, (2019)
18. Tipu Sultan, Ishak Md. Tanjinul Haque, Md. Sohanur Rahman Adnan, “*Reduction of Peak-to-Average Power Ratio (PAPR) in 5G Communication System*”, (2019)
19. Babangida Ibrahim, Md. Ekhterul Islam, Md. Selim Ahmed, “*Impact of 5G Technology on Business*”,

- (2017)
20. Md. Rasel Biswas, “*Microstrip Patch Antenna for 5G*”, (2017)
 21. Md. Sohel Rana, Samina Yesmin, “*Image Watermarking using Discrete Wavelet Transform with Identification Noise*”, (2017)
 22. A.K.M. Zilhasur Rahman, Shamsunnahar Kadir, S. M. Zahidul Islam, Md. Zahidur Rahman, “*The study of Network Virtualization & its Performance Analysis in Open-Stack Cloud Computing*”, (2017)
 23. Md. Azmal Huda, Mahmud Sayed, Md. Abu Taher, “*PAPR Reduction in OFDM Systems using Modified Gamma Correction Companding*”, (2017)
 24. Mahfuz Alam, “*Performance improvement in OFDM system by different PAPR reduction methods (PTS and SLM)*”, (2015)

Undergraduate Students

1. Md. Ripon Mia, Ahsan Jamil, Md. Sohel Ahmed, Md. Selim Reza, SK. Mushfiqul Islam, “*Campus surveillance by Drone technology*”, (2022)
2. Alok Sarker, Iftekharul Islam, Md. Emam Hossain, Mehedi Hasan, “*IOT Cloud Based Traffic Monitoring & Controlling System in Bangladesh*”, (2022)
3. Sharmeen Akter Imama, “*Implementation & analysis of 5G*”, (2021)
4. Md. Sudha Amin Badhan, “*Simulation of Digital Notice Board*”, (2021)
5. Emade Uddin Pantho, Taozia Mostafie, Tamim Islam Momin, Pronay Chandra Das, “*Study on Substation Distribution, Operation & Maintenance system of DESCO*”, (2021)
6. Md. Nazmul Kabir, Md. Ataur Rahman, Md. Asraful Islam, Amina Haque Pushpo, Arif Hossain, “*HVDC over HVAC Power Transmission System Fault Current analysis using VSC*”, (2021)
7. Md. Mirja Monir, Md. Shah Farhad, Md. Ujjal Hassan, Md. Golam Rabbani, Monira Begum, “*Installation, Testing & Commissioning of 33/11 kV & Electrical substation Equipment*”, (2021)
8. Shankar Kumar Barman, “*Impact of Regulatory Aspects on 5G Mobile Communication Systems*”, (2021)
9. Yasir Taher, “*Journey of 5G from 4G Network*”, (2018)
10. Abdullah Al-Bayezid, “*Design of a Microstrip Patch Antenna for 2.4GHz WLAN Applications*”, (2021)
11. Md. Taseen Mallick, Md. Zahid Hasan, “*Performance Analysis of Beyond 5G Compatible Downlink IBFD MIMO Wireless Communication System*”, (2020)
12. Md. Monir Hossain, “*Centralized Mobile Detector in Exam Hall*”, (2019)
13. Rokonzaman, “*Network Operation Center at Banglalion Communication Ltd.*”, (2019)
14. Md. Ajijul Hakim, Md. Nazrul Islam, “*Digital Notice Board with Finger Scanner*”, (2019)
15. Mst. Lubna Jahan, “*OOB Reduction in 5G Transmission*”, (2019)
16. Rokibul Islam, Iqbal Hossain, Mohammad Ariful Haque, Md. Emran Hossain Khan, Md. Din Islam, Md. Shahidul Islam, “*Digital Notice Board with Finger Scanner*”, (2019)
17. Mydul Islam, “*Industrial Automation of PLC Based Conveyor Belt Control System*”, (2019)
18. Md. Mahbubul Haque, Nadia Akter, “*Generator Alternator and Auto Transfer Switch ATS*”, (2019)
19. Md. Nazmul Kabir, Md. Ataur Rahman, Md. Asraful Islam, Amina Haque Pushpo, “*HVDC over HVAC Power Transmission System Faulty Current Analysis using VSC*”, (2019)
20. Md. Nasir Uddin, A.S.M Mayeedul Kabir, Md. Shakhawat Hossain Chowdhury, “*Network Operation Center: Monitoring of IPS Network*”, (2019)
21. Usaimung Murma, “*Internship on Installation, Testing & Commissioning of 33/11KV and 11/4.415KV Electrical Substation Equipment*”, (2019)
22. Md. Faruque Hossain, Md. Aslam Hossain, Md. Shamim Hossain, Md. Ashraf Siddique, “*Internship on switchgear and protective relays*”, (2019)

23. Md. Abdullah Miah, Istake Ahmed, “*Internship on GSM 325 Volvo Generator & Electrical System*”, (2019)
24. Irin Sultana Bristi, “*Internship at a Center Instructure and Development*”, (2019)
25. Shovon Roy, Md. Al Mamun Kabir, Md. Mahadi Hasan, Md. Jannatul Islam, “*Design and Implementation of a Multipurpose Robovac with Cleaning-kit*”, (2017)
26. Md. Abdullah Al Mamun, Azizul Hakim, “*Distribution & Power Transformer and Power Factor Plant*”, (2017)
27. Kazi Rezaul Hasan, Md. Ovi, Hasibul Anwar Bappy, S.M. Naimur Rahman, “*Library Book Management with RFID*”, (2017)
28. Shamar Debnath, Tareq Mahmud, Nowshin Naowar Nikita, Sayeda Rakib Hasan, “*Automatic Star Delta Starter using relays and adjustable electronic timer for Induction Motor*”, (2017)
29. Md. Rihad Hossain Hemel, Md. Murad Hosan, Chandan Datta, “*Performance of Carrier Frequency offset in OFDM system and Estimation using MATLAB*”, (2017)
30. Kazi Saiful Islam, “*Elevator Installation & Commissioning*”, (2017)
31. Taukir Ahmed, Md. Rifat Hossain, Mohammad Nurunnaee Sajib, “*Linearization Error analysis among several precoding methods in OFDM*”, (2017)
32. Md. Ziaur Rahman, Md. Mahmudul Hasan Mondal, Yusuf Md. Aminul Momenin, Md. Rezaul Karim, “*OPAMP Application Trainer Kit*”, (2017)
33. Md Abdul Kader, Abdus Salam, “*Design and Simulation of L-Slit Rectangular Microstrip Patch antenna for GSM, WLAN & WiMAX Application*”, (2016)
34. Md. Al Amin Hossain, Niaz Mohammad Hasan, Masum Billah, “*GSM Radio Network Planning & Optimization*”, (2016)
35. Ridwanul Alam, Riyad Hasan, “*Energy consumption in wireless sensor network*”, (2016)
36. M A K Wahid Mahumder, “*Secure Network Implementation with Cisco Solution*”, (2016)
37. Md. Shihabul Islam, Mst. Maliha Islam, Touhida Zaman, “*VHDL Implementation of Convolutional Encoders*”, (2016)
38. Mohammad Mahadi Hasan Foad, Fahin Raian Dip, Krishna Murari Das, “*Fast and Efficient wavelet transform for OFDM*”, (2016)
39. Md. Kamal Uddin, Safayat Ullah Zibon, Asibur Rahman, “*Operation and maintenance of GSM System*”, (2015)
40. Jilan Hossen Sohel, “*RE Engineering*”, (2015)
41. Md. Jewel Rana, “*RF optimization*”, (2015)
42. Sibbir Bin Siddique, Bidhan Sarkar, Tareq Mohammad Faruqi, “*Performance analysis & Comparison of carbon Nanotube Graphene nanoribbon & silicon nanowire Transistor*”, (2015)
43. Shakti Kumar Shaha, “*DTMF Based Home Automation System*”, (2015)
44. Md. Shohidul Islam, Md. Shafiul Bashar, Md. Minaur Rahman, Sayeduzzaman, “*Wireless rush driving detection & controlled with automatic speed breaker*”, (2015)
45. Tasnimun Tabassum, Most. Shabnaz Akter Sony, Mafiha Rahman, Mahmudul Hasan, “*Automatic Railway Gate Control*”, (2015)
46. Md. Zulfiker Ali, Shaikh Masud Alam, Md. Imroz Alam, Mohammad Syful Islam, “*Design and Simulation of a U-Shape micro strip TCH antenna of RF*”, (2015)
47. Subrata Das, Mohammad Mahfuzul Hasan Chowdhury, “*Peak to Average Power Ratios Reduction using Sinusoidal Decomposition*”, (2015)
48. Rahat Ahmed, Miti Saha, “*A New Blind Channel Estimation Technique*”, (2015)
49. Nuzahat Nurey, Sharmin Akter, Anika Tasnim, Sarmin Jahan, “*Performance analysis of FFT & wavelet based OFDM*”, (2015)
50. Avishek Batman, Tariqul Islam Pavel, Fahmida Akter, Shamsunnahar Sumi, “*EEG Signal Acquisition & Analysis*”, (2015)

51. Shuvo Saha, Md. Amadul Haque, Md. Anisur Rahman Munshi, "*Radio Frequency Optimization*", (2015)
52. S. M. Samim Reza, Md. Masud Rana, Noor Alam, Abu Khalid, "*Wireless mobile phone charger*", (2015)
53. Md. Mahdi Hasan Faisal, "*PAR Reduction in OFDM systems using Comanding*", (2015)

Appendix D

Pedagogical Training and Workshops

I strengthened my teaching practice and contributed to institutional quality assurance through active participation in pedagogical workshops at UITS. These sessions focused on curriculum design, teaching methods, assessment strategies, and outcome-based education, improving my instructional approach and enabling knowledge sharing among faculty. I also served on the Self-Assessment Committee within the IQAC Cell. The following list summarizes the key workshops and training activities that shaped my teaching development and academic leadership.

Training Attended

1. “*Inception Workshop on establishment of IQAC at UITS*” –Dr. Sanjoy Adhikar, Professor & Head, Quality Assurance Unit, HEQEP, UGC. Date: 31.05.2017
2. “*Workshop on Team Building*”, Dr. Mohammed Solaiman, Professor & Vice-Chancellor, UITS. Date: 13.06.2017
3. “*Workshop on Self-Assessment Activity*”, Dr. Mohammed Solaiman, Professor & Vice-Chancellor, UITS. Date: 24.07.2017
4. “*Workshop on Self-Assessment Survey Questionnaire*” –Dr. Shanti Narayan Ghose, Professor, Department of Business and Director, IQAC, BUBT. Date: 16.10.2017
5. “*Workshop on Curriculum Concept Model-01*”, Dr. Mohammed Solaiman, Professor & Vice-Chancellor, UITS; Muhammad J. Munir, Additional Director, IQAC, UIU; Mohammad Tohidul Islam Miya; Core Resource Person, Quality Assurance Unit, HEQEP, UGC. Date: 09.11.2017
6. “*Workshop on Curriculum Concept Model-02*”, Dr. Mohammed Solaiman, Professor & Vice-Chancellor, UITS; Dr. Khawza Iftekhar Uddin Ahmed, Professor, EEE & Director, IQAC, UIU; Dr. Md. Serajul Islam Prodhan, Associate Professor, EETE & Director, IQAC, DhIU, Date: 16.11.2017
7. “*Workshop on Curriculum Concept Model-03*”, Dr. Mohammed Solaiman, Professor & Vice-Chancellor, UITS; Dr. Nazmun Nahar, Associate Professor and Director, IQAC, NSU, Dr. Md. Mostafizer Rahman, Professor & Director, IQAC, Gono Bishwabidyalay, Savar, Dhaka. Date: 23.11.2017
8. “*Workshop on Teaching Methodology-1*”, Dr. Mohammed Solaiman, Professor & Vice-Chancellor, UITS; Dr. Muhammad Mahboob Ali, Professor, School of Economics, University of Dhaka. Date: 20.01.2018
9. “*Workshop on Teaching Methodology-2*”, Dr. Mohammed Solaiman, Professor & Vice-Chancellor, UITS; Professor Dr. M. Kaykobad, Professor, CSE, BUET. Date: 27.01.2018
10. “*Workshop on Teaching Methodology-3*”, Dr. Mohammed Solaiman, Professor & Vice-Chancellor, UITS; Dr. M.R. Kabir, Professor & Pro-Vice Chancellor, University of Asia Pacific; Dr. M. Kaykobad, Professor, CSE, BUET. Date: 03.02.2018
11. “*Workshop on Post Self-Assessment Improvement Plan (PSAIP)*”, Dr. Mahbub Ahsan Khan, Quality Assurance Specialist, QAU, HEQEP, UGC and Professor of IER at DU. Date: 30.07.2018
12. “*Team Building Workshop*”, Dr. Khawza Iftekhar Uddin Ahmed, Professor, EEE & Director, IQAC, UIU. Date: 30.10.2017

Training Conducted

1. “*SA Activity Workshop-2 (Survey Data Sharing)*”, Dr. Mohammad Mahmudul Hasan, Associate Professor, Department of EEE/ECE & SAC Member, IQAC, UITS. Date: 27.02.2018
2. “*Survey for Academic & Non-Academic Stuff, Students*”, Dr. Mohammad Mahmudul Hasan, Associate Professor, EEE/ECE & SAC Member, IQAC, UITS. Date: 4.11.17 – 13.11.17
3. “*Survey Questionnaire for Alumni*”, Dr. Mohammad Mahmudul Hasan, Associate Professor, Department of EEE/ECE & SAC Member, IQAC, UITS. Date: 08.12.2017

Appendix E

Students' Feedback

This appendix presents the end-of-course student feedback collected through the UCAM evaluation system. The survey assessed multiple aspects of the course, including course organization, course outcome achievement, learning resources, teaching quality, and assessment practices. Overall, the responses indicate a high level of student satisfaction, with most evaluation criteria receiving positive ratings and high response rates. The feedback provides an independent measure of the effectiveness of the course design, delivery, and assessment process.

Statement: The course objectives were clear.				
Response Option	Weight	Frequency	Percent	Percent Responses
Strongly Disagree	(1)	0	0.0%	
Disagree	(2)	0	0.0%	
Neutral	(3)	7	15.6%	
Agree	(4)	10	22.2%	
Strongly Agree	(5)	26	57.8%	
No Answer	(0)	2	4.4%	
				0 25 50 75 100
Response Rate		Mean	STD	Median
43/45 (95.56%)		4.44	0.76	5.0

Statement: The course objectives were clear.				
Response Option	Weight	Frequency	Percent	Percent Responses
Strongly Disagree	(1)	0	0.0%	
Disagree	(2)	3	6.7%	
Neutral	(3)	10	22.2%	
Agree	(4)	22	48.9%	
Strongly Agree	(5)	4	8.9%	
No Answer	(0)	6	13.3%	
				0 25 50 75 100
Response Rate		Mean	STD	Median
39/45 (86.67%)		3.69	0.76	4.0

Statement: The course was well organized.				
Response Option	Weight	Frequency	Percent	Percent Responses
Strongly Disagree	(1)	0	0.0%	
Disagree	(2)	0	0.0%	
Neutral	(3)	7	15.6%	
Agree	(4)	9	20.0%	
Strongly Agree	(5)	28	62.2%	
No Answer	(0)	1	2.2%	
				0 25 50 75 100
Response Rate		Mean	STD	Median
44/45 (97.78%)		4.48	0.75	5.0

Statement: Teacher discussed OBE and COs at the beginning of the semester.				
Response Option	Weight	Frequency	Percent	Percent Responses
Strongly Disagree	(1)	0	0.0%	
Disagree	(2)	0	0.0%	
Neutral	(3)	8	17.8%	
Agree	(4)	12	26.7%	
Strongly Agree	(5)	22	48.9%	
No Answer	(0)	3	6.7%	
				0 25 50 75 100
Response Rate		Mean	STD	Median
42/45 (93.33%)		4.33	0.78	5.0

Statement: Teacher explained the assessment criteria in the class.				
Response Option	Weight	Frequency	Percent	Percent Responses
Strongly Disagree	(1)	0	0.0%	
Disagree	(2)	0	0.0%	
Neutral	(3)	8	17.8%	
Agree	(4)	12	26.7%	
Strongly Agree	(5)	22	48.9%	
No Answer	(0)	3	6.7%	
				0 25 50 75 100
Response Rate		Mean	STD	Median
42/45 (93.33%)		4.33	0.78	5.0

Statement: Teacher explained how assessments align with COs.				
Response Option	Weight	Frequency	Percent	Percent Responses
Strongly Disagree	(1)	0	0.0%	
Disagree	(2)	0	0.0%	
Neutral	(3)	7	15.6%	
Agree	(4)	15	33.3%	
Strongly Agree	(5)	18	40.0%	
No Answer	(0)	5	11.1%	
				0 25 50 75 100
Response Rate		Mean	STD	Median
40/45 (88.89%)		4.28	0.74	4.0

Statement: Teacher ensured that all students achieve the COs of corresponding course.				
Response Option	Weight	Frequency	Percent	Percent Responses
Strongly Disagree	(1)	0	0.0%	
Disagree	(2)	0	0.0%	
Neutral	(3)	5	11.1%	
Agree	(4)	18	40.0%	
Strongly Agree	(5)	17	37.8%	
No Answer	(0)	5	11.1%	
				0 25 50 75 100
Response Rate		Mean	STD	Median
40/45 (88.89%)		4.30	0.68	4.0

Statement: Learning materials were relevant and useful.				
Response Option	Weight	Frequency	Percent	Percent Responses
Strongly Disagree	(1)	0	0.0%	
Disagree	(2)	0	0.0%	
Neutral	(3)	2	4.4%	
Agree	(4)	18	40.0%	
Strongly Agree	(5)	23	51.1%	
No Answer	(0)	2	4.4%	
				0 25 50 75 100
Response Rate		Mean	STD	Median
43/45 (95.56%)		4.49	0.59	5.0

Statement: Recommended readings, books etc. were relevant and appropriate.				
Response Option	Weight	Frequency	Percent	Percent Responses
Strongly Disagree	(1)	0	0.0%	
Disagree	(2)	0	0.0%	
Neutral	(3)	5	11.1%	
Agree	(4)	12	26.7%	
Strongly Agree	(5)	26	57.8%	
No Answer	(0)	2	4.4%	
				0 25 50 75 100
Response Rate		Mean	STD	Median
43/45 (95.56%)		4.49	0.69	5.0

Statement: Facilities of the classroom were adequate.				
Response Option	Weight	Frequency	Percent	Percent Responses
Strongly Disagree	(1)	0	0.0%	
Disagree	(2)	3	6.7%	
Neutral	(3)	7	15.6%	
Agree	(4)	18	40.0%	
Strongly Agree	(5)	9	20.0%	
No Answer	(0)	8	17.8%	
				0 25 50 75 100
Response Rate		Mean	STD	Median
37/45 (82.22%)		3.89	0.86	4.0

Statement: The overall environment in the classroom was conducive to learning.				
Response Option	Weight	Frequency	Percent	Percent Responses
Strongly Disagree	(1)	0	0.0%	
Disagree	(2)	0	0.0%	
Neutral	(3)	8	17.8%	
Agree	(4)	15	33.3%	
Strongly Agree	(5)	19	42.2%	
No Answer	(0)	3	6.7%	
				0 25 50 75 100
Response Rate		Mean	STD	Median
42/45 (93.33%)		4.26	0.76	4.0

Statement: Course was well structured to achieve learning outcomes.				
Response Option	Weight	Frequency	Percent	Percent Responses
Strongly Disagree	(1)	0	0.0%	
Disagree	(2)	2	4.4%	
Neutral	(3)	5	11.1%	
Agree	(4)	22	48.9%	
Strongly Agree	(5)	9	20.0%	
No Answer	(0)	7	15.6%	
				0 25 50 75 100
Response Rate		Mean	STD	Median
38/45 (84.44%)		4.00	0.76	4.0

Statement: Teacher has adequate knowledge on the course.				
Response Option	Weight	Frequency	Percent	Percent Responses
Strongly Disagree	(1)	0	0.0%	
Disagree	(2)	0	0.0%	
Neutral	(3)	0	0.0%	
Agree	(4)	15	33.3%	
Strongly Agree	(5)	30	66.7%	
No Answer	(0)	0	0.0%	
				0 25 50 75 100
Response Rate		Mean	STD	Median
45/45 (100.00%)		4.67	0.47	5.0

Statement: Teacher was regular in class.				
Response Option	Weight	Frequency	Percent	Percent Responses
Strongly Disagree	(1)	0	0.0%	
Disagree	(2)	0	0.0%	
Neutral	(3)	0	0.0%	
Agree	(4)	15	33.3%	
Strongly Agree	(5)	27	60.0%	
No Answer	(0)	3	6.7%	
				0 25 50 75 100
Response Rate		Mean	STD	Median
42/45 (93.33%)		4.64	0.48	5.0

Statement: Ideas and concepts were presented clearly.				
Response Option	Weight	Frequency	Percent	Percent Responses
Strongly Disagree	(1)	0	0.0%	
Disagree	(2)	0	0.0%	
Neutral	(3)	2	4.4%	
Agree	(4)	17	37.8%	
Strongly Agree	(5)	25	55.6%	
No Answer	(0)	1	2.2%	
				0 25 50 75 100
Response Rate		Mean	STD	Median
44/45 (97.78%)		4.52	0.58	5.0

Statement: Teacher was dedicated to improving teaching method.				
Response Option	Weight	Frequency	Percent	Percent Responses
Strongly Disagree	(1)	0	0.0%	
Disagree	(2)	0	0.0%	
Neutral	(3)	4	8.9%	
Agree	(4)	18	40.0%	
Strongly Agree	(5)	20	44.4%	
No Answer	(0)	3	6.7%	
				0 25 50 75 100
Response Rate		Mean	STD	Median
42/45 (93.33%)		4.38	0.65	4.0

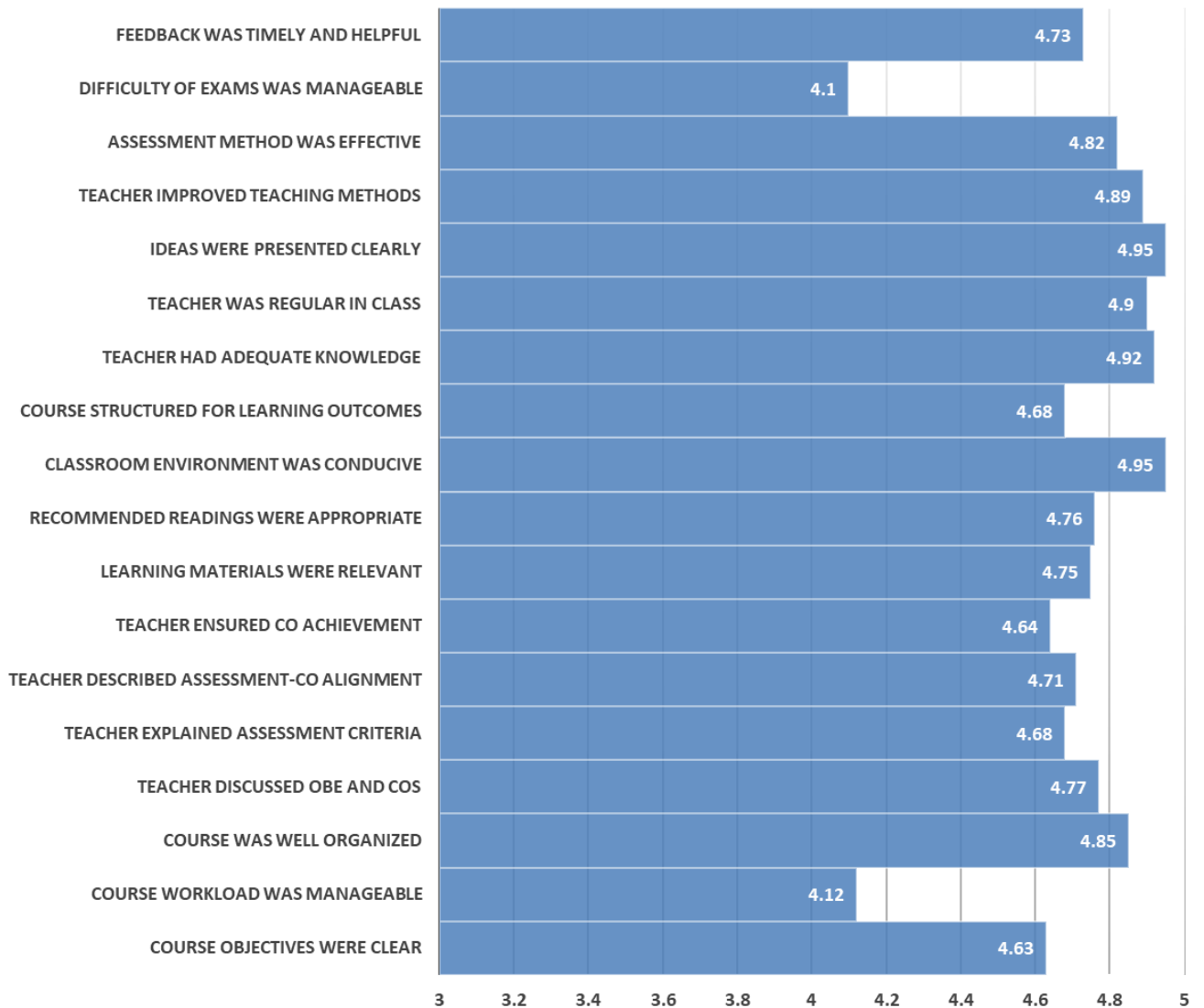
Assessment

Statement: The method of assessment was effective.				
Response Option	Weight	Frequency	Percent	Percent Responses
Strongly Disagree	(1)	0	0.0%	
Disagree	(2)	0	0.0%	
Neutral	(3)	6	13.3%	
Agree	(4)	12	26.7%	
Strongly Agree	(5)	22	48.9%	
No Answer	(0)	5	11.1%	
				0 25 50 75 100
Response Rate		Mean	STD	Median
40/45 (88.89%)		4.40	0.73	5.0

Statement: Difficulty of exam questions was manageable.				
Response Option	Weight	Frequency	Percent	Percent Responses
Strongly Disagree	(1)	0	0.0%	
Disagree	(2)	0	0.0%	
Neutral	(3)	2	4.4%	
Agree	(4)	16	35.6%	
Strongly Agree	(5)	24	53.3%	
No Answer	(0)	3	6.7%	
				0 25 50 75 100
Response Rate		Mean	STD	Median
42/45 (93.33%)		4.52	0.59	5.0

Statement: Feedback on assessment was timely and helpful.				
Response Option	Weight	Frequency	Percent	Percent Responses
Strongly Disagree	(1)	0	0.0%	
Disagree	(2)	0	0.0%	
Neutral	(3)	1	2.2%	
Agree	(4)	17	37.8%	
Strongly Agree	(5)	22	48.9%	
No Answer	(0)	5	11.1%	
				0 25 50 75 100
Response Rate		Mean	STD	Median
40/45 (88.89%)		4.53	0.55	5.0

Student Feedback on Course EEE 315 Digital Signal Processing



Additional Comments

Course Content and Organization

Topics were clear and well-paced, flow was easy to follow, weekly outlines helped, slides were concise, balanced theory and practice, smooth progression overall, but some chapters felt rushed, a few sections overloaded, some content outdated, could use more real-life examples and case studies, mid-course workload high in parts, more practical exercises would help, clearer links between topics needed, some repetition could be reduced, objectives for each week could be highlighted.

Course Outcome (CO) Achievement

COs were explained clearly, teacher gave good overview of OBE, assessment links were helpful, teacher pushed us to achieve COs, clear start-of-semester briefing, regular reinforcement was nice, mapping examples were useful, transparent criteria, overall CO direction clear, but sometimes CO links were hard to see, a few COs seemed overlapping, marks-to-CO mapping unclear, some CO targets felt high, more feedback on individual CO progress needed, reminders of COs during semester would help, better check-ins to track CO achievement suggested.

Learning Resources and Learning Environment

Learning materials were useful and relevant, reading list solid, classroom comfortable, multimedia used well, handouts helpful, resources well organized, conducive environment, adequate facilities, clean classroom, reference materials helpful, slides mostly clear, but some materials outdated, more practice problems wanted, readings sometimes too advanced, a few broken links, more visual aids would help, Wi-Fi and access to resources could be improved, lighting and air quality sometimes poor, solutions to exercises needed.

Teaching Method and Teaching Quality

Teacher was knowledgeable and dedicated, explanations mostly clear, concepts broken down well, class structured to achieve learning outcomes, engaging style, regular in class, encouraged participation, improved teaching over semester, committed, good pacing overall, but sometimes lectures too fast, needed more interactive sessions and group work, more examples and case studies would help, check understanding regularly, reduce repetition, clarify assignment instructions, provide summaries at end of class, more visuals and demos suggested, adapt pace to student comprehension.

Assessment

Assessment methods mostly fair, feedback helpful and timely, exam difficulty manageable, good variety of questions, clear marking criteria, aligned with learning outcomes, some tricky questions, partial credit could help, more formative assessments for practice, better alignment with taught content, revision sessions before exams would help, marking rubrics sometimes unclear, instructions for tasks could be clearer, some exams felt heavy, preparation tips appreciated, overall feedback quality strong.

University of Information Technology & Sciences (UITS)

Department of Electrical and Electronic Engineering

Course Title: Digital Signal Processing

Course Code: EEE 315, Credit: 3.00

Faculty Name: Mohammad Mahmudul Hasan

STUDENT FEEDBACK FORM

(to be filled up by each student at the time of course completion)

Evaluate the following aspects of the course by marking “✓” in box of corresponding column according to the scale given:

5- Strongly agree, 4- Agree, 3- Neutral, 2- Disagree, 1- Strongly disagree

A. Course Content and Organization	5	4	3	2	1
1. The course objectives were clear.	✓				
2. The course workload was manageable.		✓			
3. The course was well organized.	✓				
4. Additional Comments: <i>Explanations clear, examples useful, pace sometimes fast</i>					

B. Course Outcome (CO) Achievement	5	4	3	2	1
1. Teacher discussed OBE and COs at the beginning of the semester.	✓				
2. Teacher explained the assessment criteria in the class.	✓				
3. Teacher described the relationship between assessments and COs to achieve the course outcomes.	✓				
4. Teacher worked hard to ensure that all students achieve the COs of corresponding course.	✓				
5. Additional Comments: <i>CO recap during assessments would help clarity</i>					

C. Learning Resources and Learning Environment	5	4	3	2	1
1. Learning materials were relevant and useful.	✓				
2. Recommended readings, books etc. were relevant and appropriate.	✓				
3. The overall environment in the classroom was conducive to learning.	✓				
4. Facilities of the classroom were adequate.			✓		
5. Additional Comments: <i>Facilities were adequate, projector could be sharper</i>					

D. Teaching Method and Teaching Quality	5	4	3	2	1
1. Course was well structured to achieve learning outcomes.	✓				
2. Teacher has adequate knowledge on the course.	✓				
3. Teacher was regular in class.	✓				
4. Ideas and concepts were presented clearly.	✓				
5. Teacher was dedicated to improving teaching method.	✓				
6. Additional Comments:	He's always on time, pays attention to us, inspires us				

E. Assessment	5	4	3	2	1
1. The method of assessment was effective.	✓				
2. Difficulty of exam questions was manageable.		✓			
3. Feedback on assessment was timely and helpful.	✓				
4. Additional Comments:	Consistent, transparent, helped identify weak spots				

F. Overall Evaluation

- The best features of the course were:

clear explanations, structured lectures, good hands-on, real-world examples, supportive guidance, well paced content, organized assessments. Engineering is fun finally ...

- The course could have been improved by:

A few more practice sets, occasional recap sessions

References

- Abramczyk, Anna and Susanne Jurkowski (May 2020). "Cooperative learning as an evidence-based teaching strategy: what teachers know, believe, and how they use it". In: *Journal of Education for Teaching* 46.3, pp. 296–308.
- Adams, Nancy E (2015). "Bloom's taxonomy of cognitive learning objectives". In: *Journal of the Medical Library Association : JMLA* 103.3.
- Al Abri, Jawaher Said and Abdo Mohammed Al-Mekhlafi (Dec. 2024). "Think-Pair-Share: An Active Learning Strategy to Enhance EFL Learners' Oral Communication Skills". In: *World Journal of English Language* 15.3, p. 165.
- Aldridge, Jill M (2012). "Outcomes-Focused Learning Environments". In: *Second International Handbook of Science Education*.
- Armbruster, Peter et al. (Sept. 2009). "Active learning and student-centered pedagogy improve student attitudes and performance in introductory biology". In: *CBE Life Sciences Education* 8.3, pp. 203–213.
- Bansal, Seema et al. (May 2019). "Collaborative research in modern era: Need and challenges". In: *Indian Journal of Pharmacology* 51.3, p. 137.
- Biggs, John B and Catherine Tang (2011). *Teaching for Quality Learning at University*. 4th Ed. Buckingham, England: Open University Press.
- Bonwell, Charles C. and James A. Eison (1991). "Active Learning: Creating Excitement in the Classroom (J-B ASHE Higher Education Report Series (AEHE))". In: *AAHE Bulletin*, p. 128.
- Capone, Roberto (Mar. 2022). "Blended Learning and Student-centered Active Learning Environment: a Case Study with STEM Undergraduate Students". In: *Canadian Journal of Science, Mathematics and Technology Education* 2022 22:1 22.1, pp. 210–236.
- Clapper, Timothy C (2010). "Creating the safe learning environment." In: *PAILAL* 3.2, pp. 1–6.
- Das, Sibadatta et al. (Oct. 2021). "Case-based learning: Modern teaching tool meant for present curriculum: A behavioral analysis from faculties' perspective". In: *Journal of Education and Health Promotion* 10.1.
- Eison, James (Jan. 1991). "Active Learning: Creating Excitement in the Classroom. ERIC Digest". In: *Ashe Eric Higher Education Report*.
- Ferreira, Juliene Madureira, Kateryna Zabolotna, and Samin Lee (Nov. 2025). "Teaching twenty-first-century skills: examining collaborative learning in initial teacher education in Finnish universities". In: *Scandinavian Journal of Educational Research* 69.7, pp. 1319–1340.
- Genn, J. M. (2001a). "AMEE Medical Education Guide No. 23 (Part 1): Curriculum, environment, climate, quality and change in medical education—a unifying perspective". In: *Medical Teacher* 23.4, pp. 337–344.
- Genn, J. M. (2001b). "AMEE Medical Education Guide No. 23 (Part 2): Curriculum, environment, climate, quality and change in medical education – a unifying perspective". In: *Medical Teacher* 23.5, pp. 445–454.
- Gokhale, A A (1995a). "Collaborative Learning Enhances Critical Thinking". In: *Journal of Technology Education* 7, pp. 22–30.
- Gokhale, A A (1995b). "Collaborative Learning Enhances Critical Thinking". In: *Journal of Technology Education* 7, pp. 22–30.
- Guenther, Amy R and Cathleen M Abbott (2024). "Think-Pair-Share: Promoting Equitable Participation and In-Depth Discussion". In: *PRiMER (Leawood, Kan.)* 8, p. 7.
- Hutchinson, Linda (2003). "Educational environment". In: *BMJ* 326.7393.
- Johnson, David W and Roger T Johnson (1999). "Making cooperative learning work". In: *Theory Into Practice* 38.2.
- Katawazai, Rahmatullah (May 2021). "Implementing outcome-based education and student-centered learning in Afghan public universities: the current practices and challenges". In: *Heliyon* 7.5.

- Kaur, Gurleen et al. (Jan. 2020). "Case-Based Learning as an Effective Tool in Teaching Pharmacology to Undergraduate Medical Students in a Large Group Setting". In: *Journal of Medical Education and Curricular Development* 7.
- Laal, Marjan and Seyed Mohammad Ghodsi (2012). "Benefits of collaborative learning". In: *Procedia - Social and Behavioral Sciences* 31.
- Li, Hong (2025). "Impact of collaborative learning on student engagement in college English programs: mediating effect of peer support and moderating role of group size". In: *Frontiers in Psychology* 16, p. 1525192.
- Mahbubul Syeed, M. M. et al. (2022). "Outcome Based Education (OBE): Defining the Process and Practice for Engineering Education". In: *IEEE Access* 10, pp. 119170–119192.
- Maor, Rotem et al. (Sept. 2025). "Teaching creatively and teaching for creativity – theory, teachers' attitudes, and creativity-based practices". In: *Educational Studies* 51.5, pp. 958–972.
- Mazur, Eric (2009). "Farewell, Lecture?" In: *Science* 323.5910.
- Noushad, P P (2024). "Introduction to Outcome-Based Education". In: *Springer Texts in Education*.
- Olewnik, Andrew, Yunjeong Chang, and Mengchen Su (Apr. 2023). "Co-curricular engagement among engineering undergrads: do they have the time and motivation?" In: *International Journal of STEM Education* 2023 10:1 10.1, pp. 27–.
- Prince, Michael (July 2004). "Does active learning work? A review of the research". In: *Journal of Engineering Education* 93.3, pp. 223–231.
- Rao, N J and N J Rao (n.d.). "Outcome-based Education: An Outline". In: *Higher Education for the Future* 7.1 ().
- Rathy, G. A., P. Sivasankar, and T. G. Gnanasambandhan (Jan. 2020). "Developing a knowledge structure using Outcome based Education in Power Electronics Engineering". In: *Procedia Computer Science* 172, pp. 1026–1032.
- Sunra, La, Nur Aeni, and Fauzan Hari Sudding Sally (Jan. 2024). "A Comprehensive Exploration of Outcome-Based Education Principles and Practices". In: *Asian Journal of Education and Social Studies* 50.1, pp. 1–9.
- Thompson, Nancy E. and Julie P. Wheeler (Sept. 2008). "Learning Environment: Creating and Implementing a Safe, Supportive Learning Environment." In: *Journal of Family & Consumer Sciences Education* 26.2, p. 33.
- West, Joyce (Jan. 2023). "Utilizing Bloom's taxonomy and authentic learning principles to promote preservice teachers' pedagogical content knowledge". In: *Social Sciences & Humanities Open* 8.1, p. 100620.
- Widiana, I. Wayan et al. (Dec. 2023). "Bloom's revised taxonomy-oriented learning activity to improve reading interest and creative thinking skills". In: *Cogent Education* 10.2.
- Yang, Xigui (Jan. 2023). "A Historical Review of Collaborative Learning and Cooperative Learning". In: *TechTrends* 2023 67:4 67.4, pp. 718–728.
- Young, J. E., M. I. Williamson, and T. G. Egan (May 2015). "Students' reflections on the relationships between safe learning environments, learning challenge and positive experiences of learning in a simulated GP clinic". In: *Advances in Health Sciences Education* 2015 21:1 21.1, pp. 63–77.